

CyAbrB2 is a nucleoid-associated protein in *Synechocystis* controlling hydrogenase expression during fermentation

Reviewed Preprint

v2 • July 10, 2024

Revised by authors

Reviewed Preprint

v1 • February 8, 2024

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Abstract

The *hox* operon in *Synechocystis* sp. PCC 6803, encoding bidirectional hydrogenase responsible for H₂ production, is transcriptionally upregulated under microoxic conditions. Although several regulators for *hox* transcription have been identified, their dynamics and higher-order DNA structure of *hox* region in microoxic conditions remain elusive. We focused on key regulators for the *hox* operon: cyAbrB2, a conserved regulator in cyanobacteria, and SigE, an alternative sigma factor. Chromatin immunoprecipitation-sequencing revealed that cyAbrB2 binds to the *hox* promoter region under aerobic conditions, with its binding being flattened in microoxic conditions. Concurrently, SigE exhibited increased localization to the *hox* promoter under microoxic conditions. Genome-wide analysis revealed that cyAbrB2 binds broadly to AT-rich genome regions and represses gene expression. Moreover, we demonstrated the physical interactions of the *hox* promoter region with its distal genomic loci. Both the transition to microoxic conditions and the absence of cyAbrB2 influenced the chromosomal interaction. From these results, we propose that cyAbrB2 is a cyanobacterial nucleoid-associated protein (NAP), modulating chromosomal conformation, which blocks RNA polymerase from the *hox* promoter in aerobic conditions. We further infer that cyAbrB2, with altered localization pattern upon microoxic conditions, modifies chromosomal conformation in microoxic conditions, which allows SigE-containing RNA polymerase to access the *hox* promoter. The coordinated actions of this NAP and the alternative sigma factor are crucial for the proper *hox* expression in microoxic conditions. Our results highlight the impact of cyanobacterial chromosome conformation and NAPs on transcription, which have been insufficiently investigated.

eLife assessment

The authors provide **solid** data on a functional investigation of potential nucleoid-associated proteins and the modulation of chromosomal conformation in a model cyanobacterium. These **valuable** findings will be of interest to the chromosome and microbiology fields. Additional analysis and the tempering of conclusions has helped to improve the work, although further refinement remains possible.

<https://doi.org/10.7554/eLife.94245.2.sa2>

Introduction

Cyanobacteria perform fermentation, using glycolytic products as electron acceptors. (1 [↗](#)). Cyanobacteria have multiple fermentation pathways according to the environment. For example, the freshwater living cyanobacterium *Synechocystis* sp. PCC 6803 (hereafter referred to as *Synechocystis*) produces acetate, lactate, dicarboxylic acids, and hydrogen (1 [↗](#), 2 [↗](#)).

Hydrogen is generated in quantities comparable to lactate and dicarboxylic acids as the result of electron acceptance in the dark microoxic condition (3 [↗](#), 4 [↗](#)). Bidirectional hydrogenase is a key enzyme for H₂ production from protons (5 [↗](#)) and is commonly found in cyanobacteria (6 [↗](#)). Cyanobacterial hydrogenase comprises five subunits (HoxEFUHY) containing nickel and Fe-S clusters (7 [↗](#)). This enzyme can utilize NADH, reduced ferredoxin, and flavodoxin as substrates (8 [↗](#)). Hydrogenase mainly receives reduced ferredoxin from pyruvate-ferredoxin oxidoreductase (PFOR) in the microoxic condition (8 [↗](#), 9 [↗](#)). Although hydrogenase and PFOR are O₂ sensitive, they can work under aerobic conditions (10 [↗](#)–12 [↗](#)). Therefore, uncontrolled expression of *hox* operon and *nifH* (coding gene of PFOR) may hamper metabolism under photosynthetic conditions. Furthermore, genetic manipulations on *Synechocystis* have demonstrated that modulating the expression of certain enzymes including hydrogenase can alter fermentative metabolic flow (3 [↗](#), 4 [↗](#), 13 [↗](#)). This provides evidence that transcription regulates the fermentative pathway. Thus, transcriptional regulation in response to the environment is essential for optimal energy cost performance.

Promoter recognition by RNA polymerases is an essential step in transcriptional regulation. Sigma factors, subunits of RNA polymerase, recognize core promoter sequences. Transcription factors can also bind to promoter regions to suppress or promote RNA polymerase transcription. As well as recruitment or blocking of RNA polymerase, some transcriptional factors, known as Nucleoid-associated proteins (NAPs), modulate chromosomal conformation to regulate transcription (14 [↗](#)). NAPs are common in bacteria, but cyanobacterial NAPs remain unidentified, and higher-order DNA structure in cyanobacteria is yet to be shown. A recent study suggested that the manipulation of chromosomal supercoiling impacts transcriptional properties in cyanobacteria (15 [↗](#)). There is room for consideration of NAPs modulating chromosomal conformation and regulating expression in cyanobacteria.

In *Synechocystis*, the coding genes of HoxEFUHY form a single operon (sll1220–1226), while PFOR is encoded in the *nifH* (sll0741) gene. Both *hox* and *nifH* operons are highly expressed under microoxic conditions (16 [↗](#)). Genetic analysis has revealed that multiple global transcriptional regulators control *hox* and *nifH* expression. Sigma factor SigE (Sll1689) promotes the expression of *hox* and *nifH* operons (17 [↗](#), 18 [↗](#)), while transcription factor cyAbrB2 (Sll0822) represses them (19 [↗](#), 20 [↗](#)). Positive regulators for the *hox* operon include LexA (Sll1626) and cyAbrB1 (Sll0359) (21 [↗](#)–23 [↗](#)).

SigE, an alternative sigma factor, controls the expression of genes involved in glycogen catabolism and glycolysis, as well as PFOR/*nifH* and hydrogenase (17 [↗](#)). SigE shows a high amino acid sequence similarity with the primary sigma factor SigA, which is responsible for transcribing housekeeping and photosynthetic genes (24 [↗](#)). A ChIP-seq study revealed that, while most SigE-binding sites are the same as SigA, SigE exclusively occupies the promoters of glycogen catabolism and glycolysis (25 [↗](#)).

CyAbrB2 and its homolog cyAbrB1 are transcription factors highly conserved in cyanobacteria. For example, cyAbrB homologs in *Anabaena* sp. PCC7120 is involved in heterocyst formation (26 [↗](#)). CyAbrB2 in *Synechocystis* regulates the expression of several genes involved in carbon metabolism, nitrogen metabolism, and cell division (20 [↗](#), 27 [↗](#), 28 [↗](#)). CyAbrB2 binds to the *hox*

promoter *in vitro* and represses its expression *in vivo* (19 [↗](#)). CyAbrB1, an essential gene, physically interacts with the cyAbrB2 protein (29 [↗](#)) and binds the *hox* promoter region *in vitro* to promote its expression (21 [↗](#)).

To explore the dynamics of those transcription factors governing the expression of *hox* operon, we conducted a time-course analysis of the transcriptome and ChIP-seq of SigE and cyAbrB2. Our ChIP-seq and transcriptome analysis showed the NAPs-like nature of cyAbrB2, which prompted us to conduct a chromosomal conformation capture assay. 3C analysis explored the physical interaction between the *hox* promoter region and its downstream and upstream genomic region in the aerobic condition, and some loci changed interaction frequency upon entry to the microoxic condition. Furthermore, some interactions in the $\Delta cyabrB2$ mutant were different from those of the wildtype. From those experiments, we propose that cyAbrB2 modulates chromosomal conformation like NAPs, allowing access to the SigE-containing RNA polymerase complex on the *hox* promoter, by which the *hox* operon achieves distinct expression dynamics. Chromosomal conformation of bacteria is a growing area of interest, and the findings of this study have brought insight into the transcriptional regulation of cyanobacteria.

Results

Transcriptomes on entry to dark microoxic conditions

To understand transcriptional regulation under microoxic conditions, we conducted a time-course transcriptome capturing light aerobic and dark microoxic conditions at 1-, 2-, and 4-hours time points (Figure 1A [↗](#)). Gene set enrichment analysis based on KEGG pathway revealed that many biological pathways, including photosynthesis and respiration (oxidative phosphorylation), were downregulated by the transition to dark microoxic conditions from light aerobic conditions (Figure 1B [↗](#)). Upregulated pathways included butanoate metabolism and two-component systems. The enrichment in the butanoate metabolism pathway indicates the upregulation of genes involved in carbohydrate metabolism. We further classified genes according to their expression dynamics. Within 1 hour of switching from aerobic to microoxic conditions, the expression levels of 508 genes increased more than 2-fold. Furthermore, genes with increased expression levels were classified into four groups based on the time course (Figures 1C [↗](#) and S1 [↗](#)). Of the 508 genes, 28 were termed “transiently upregulated genes” due to their decreased expression upon the comparison of 1- and 4-hours incubation under microoxic conditions ($\log_2$ fold change < -0.5), and 119 were termed “continuously upregulated genes,” which continuously increased between 1- and 4-hours incubation under microoxic conditions ($\log_2$ -fold change > 0.5). Other than 508 genes 2-fold upregulated within 1 hour, 28 genes showed more than 4-fold increment within 4 hours but not upregulated within 1 hour. We combined those “Late upregulated genes” with 508 genes and referred to as “All upregulated genes” in the subsequent analysis (Figure S1 [↗](#)). Mapping the classified genes to central carbon metabolism revealed that *nifH* encoding PFOR and *hox* operon encoding a bidirectional hydrogenase complex were transiently upregulated (Figure 1D [↗](#) and Table 1 [↗](#)). RTqPCR verified the transient expression of *hoxF*, *hoxH*, and *nifH* (Figure S2 [↗](#)).

SigE and cyAbrB2 control the expression of transiently upregulated genes

The functional correlation between hydrogenase and PFOR, encoded by the *hox* operon and *nifH*, suggests that transient upregulation has physiological significance. We focused on transiently upregulated genes and attempted to reveal the regulatory mechanism underlying transient upregulation. While SigE promotes the expression of *hox* and *nifH*, cyAbrB2 represses them (17 [↗](#), 19 [↗](#), 20 [↗](#)). We confirmed that the deletion of *sigE* and *cyabrB2* ($\Delta sigE$ and $\Delta cyabrB2$, respectively)

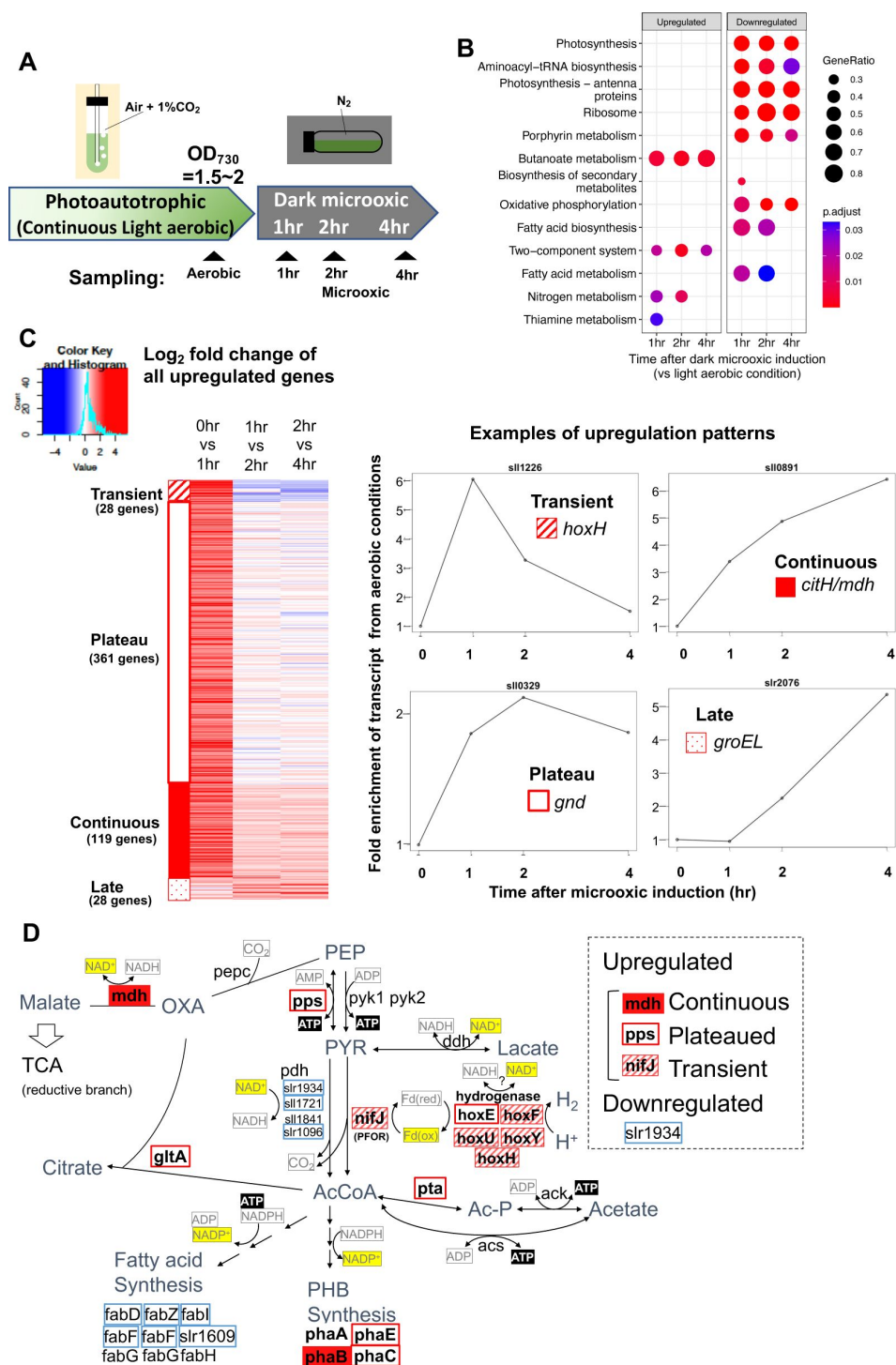


Figure 1

(A) Schematic diagram for the sampling of cells under aerobic and microoxic conditions. (B) Gene set enrichment analysis on time-course transcriptome data. KEGG pathways enriched in upregulated or downregulated genes after 1-, 2-, and 4-h incubation under microoxic conditions are shown. (C) (left) Heatmap showing expression change in all upregulated genes over the time-course. Genes classified into Transient (striped square), Plateau (open square), Continuous (filled square), and Late (dotted square) were clustered into subgroups and sorted by the gene name. (right) Examples of genes are classified into each expression pattern. (D) The classified genes were mapped to central carbon metabolism, centered on pyruvate. PEP: phosphoenolpyruvate, PYR: pyruvate, AcCoA: acetyl CoA, Ac-P: acetyl phosphate, OXA: oxaloacetate, PHB: polyhydroxy butyrate. TCA: tricarboxylic acid cycle.

		Operon
Oxidoreductase		
sll0741	<i>nifJ</i> /"pyruvate-ferredoxin/flavodoxin oxidoreductase"	TU3296
sll0743	hypothetical protein	
sll0744	dihydroorotate dehydrogenase (fumarate)	
sll1221	<i>hoxF</i> /"bidirectional [NiFe] hydrogenase diaphorase subunit"	TU1714
sll1222	unknown protein	
sll1223	<i>hoxU</i> /"bidirectional [NiFe] hydrogenase diaphorase subunit"	
sll1224	<i>hoxY</i> /"NAD-reducing hydrogenase small subunit"	
sll1225	unknown protein	
sll1226	<i>hoxH</i> /"NAD-reducing hydrogenase large subunit"	TU1089
slr1434	<i>pntB</i> /"H ⁺ -translocating NAD(P) transhydrogenase subunit beta"	
Transporter		
sll1450	<i>nrtA</i> /"nitrate/nitrite transport system substrate-binding protein"	TU1023
sll1451	<i>nrtB</i> /"nitrate/nitrite transport system permease protein"	
sll1452	<i>nrtC</i> /"nitrate/nitrite transport system ATP-binding protein"	
sll1453	<i>nrtD</i> /"nitrate/nitrite transport system ATP-binding protein"	
Two-component system		
slr1214	twitching motility two-component system response regulator PilG	TU905
slr1215	unknown protein	TU907
Glycosyl transferase		
slr2116	<i>spsA</i> /"spore coat polysaccharide biosynthesis protein; SpsA"	TU1673

Table 1

List of transiently upregulated genes

Protease		
sll1009	<i>frpC</i> /"iron-regulated protein"	TU491
Insertion sequence (Transposase)		
slr1523	transposase	TU1659
sll1985	transposase	TU1589
sll7001	transposase	NA
sll7003	toxin FitB	TU7001
ssl0172	transposase	TU3163
Other		
slr1260	hypothetical protein	TU1446
slr0668	unknown protein	TU3532
slr5127	unknown protein	TU5127
sll0710	unknown protein	TU97
sll1307	unknown protein	TU1224

The list of transiently upregulated genes was merged by transcriptional units and sorted by function. The

transcriptional unit information was obtained from a previous study (Kopf et al., 2014).

Table 1 (continued)

affected the expression of *hoxF*, *hoxH*, and *niff* by RT-qPCR (Figure S1). Thus, we conducted a time-course transcriptome analysis of $\Delta sigE$ and $\Delta cyabr2$ under aerobic conditions and after 1- and 2-hours cultivation in dark microoxic conditions (Figures 2A and S3). The transcriptome data confirmed that SigE and cyAbrB2 regulated *hox* operon expression (Figure 2B). At each time point, we searched for differentially expressed genes (DEGs) between mutants and wildtype with a more than 2-fold expression change and a false discovery rate (FDR) less than 0.05. We found that deleting *sigE* or *cyabr2* preferentially affected the expression of transiently upregulated genes, not limited to *hox* and *niff* operons (Figures 2C and 2D). Interestingly, *cyabr2* deletion resulted in the upregulated expression of transient genes under aerobic conditions, in contrast to 1-hour cultivation under microoxic conditions (Figure 2C).

Genome-wide analysis of cyAbrB2, cyAbrB1, and SigE via ChIP-seq

To decipher the regulatory mechanism of transiently upregulated genes, we must first comprehend the fundamental features and functions of these transcriptional regulators. Therefore, a genome-wide survey of cyAbrB2 and SigE occupation (Figure S4) combined with transcriptome data was done. Specifically, we generated a *Synechocystis* strain in which cyAbrB2 was epitope-tagged and performed a ChIP-seq assay under aerobic and microoxic conditions (Figure S4A). SigE-tagged strains previously constructed and analyzed elsewhere were also employed (25). The primary sigma factor SigA was also analyzed to determine SigE-specific binding. In addition to cyAbrB2, we tagged and analyzed cyAbrB1, which is the interactor of cyAbrB2 and positively regulates the *hox* operon.

CyAbrB2 binds to long tracts of the genomic region and suppresses genes in the binding region

The ChIP-seq data showed that cyAbrB2 bound to long tracts of the genomic region with lower GC content than the whole genome *Synechocystis* (Figures 3A and 3B). Vice versa, regions exhibiting lower GC contents showed a greater binding signal of cyAbrB2 (Figure 3C). This correlation was not a systematic bias of next-generation sequencing because the binding signals of SigE, SigA, and control showed no negative correlation to GC contents (Figures S5A and S5B). The binding regions of cyAbrB2 called by peak-caller covered 15.7% of the entire genome length. 805 of 3614 genes overlapped with cyAbrB2 binding regions, and almost half (399 of 805 genes) were entirely covered by cyAbrB2 binding regions. The cyAbrB2 binding regions included 80 of 125 insertion sequence elements (Figure 3D). Comparison with the transcriptome of $\Delta cyabr2$ revealed that cyAbrB2 tended to suppress the genes overlapping with its binding regions under aerobic conditions (Figures 3A and 3E). A survey of the genomic localization of cyAbrB1 under aerobic conditions revealed that cyAbrB1 and cyAbrB2 shared similar binding patterns (Figures 3A and S5C). Due to the essentiality of cyAbrB1, we did not perform transcriptome analysis on a cyAbrB1-disrupted strain. Instead, we referred to the recent study performing transcriptome of cyAbrB1 knockdown strain, whose cyAbrB1 protein amount drops by half (30). Among 24 genes induced by cyAbrB1 knockdown, 12 genes are differentially downregulated genes in *cyabr2Δ in our study (Figure S5D).*

CyAbrB2 binds to transiently upregulated genes

The binding regions of cyAbrB2 overlapped 17 of 28 transiently upregulated genes, showing significant enrichment from all upregulated genes (Figure 4A). The transiently upregulated genes belong to 17 transcriptional units (TUs), according to the previous study (31), and cyAbrB2 tends to bind TUs with transiently upregulated genes (Figure 4B). While cyAbrB2 covered the entire length of insertion sequences and unknown proteins, its binding positions on other transient genes were diverse (Figure 4C). Specifically, the *hox* and *niff* operons had two distinct

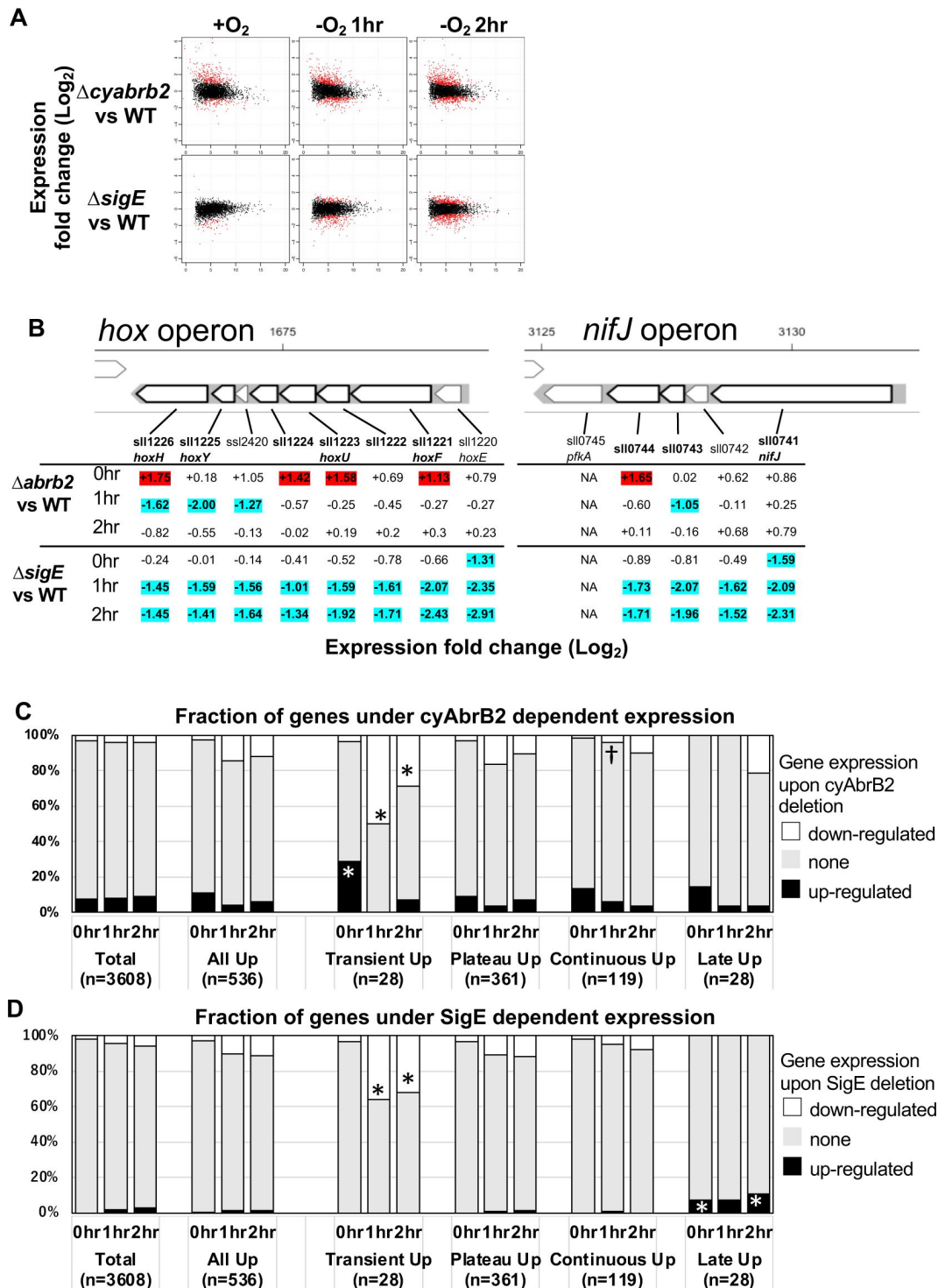


Figure 2

(A) MA plot showing fold change (y-axis) and average (x-axis) of gene expression between wildtype and mutant strains at each timepoint. Red dots indicate defined DEGs ($|\log_2 FC| > 1$ with FDR < 0.05). (B) Log2 scaled expression fold change in genes in the *hox* and *nifJ* operons upon $\Delta cyabr2$ and $\Delta sigE$ under aerobic conditions (0 hr), 1 hour after microoxic condition (1 hr), and 2 hours after microoxic condition (2 hr). DEGs are marked with sky blue (downregulated upon deletion) or red (upregulated upon deletion). (C and D) Fraction of upregulated and downregulated genes upon the (C) $\Delta cyabr2$ and (D) $\Delta sigE$ at the timepoints of aerobic conditions (0 hr), 1 hour after anoxic condition (1 hr), and 2 hour after anoxic condition (2 hr). Genes are classified according to [Figure 1C](#). Asterisk (*) and dagger (†) denote statistically significant enrichment and anti-enrichment compared with all upregulated genes tested by multiple comparisons of Fisher's exact test (FDR < 0.05).

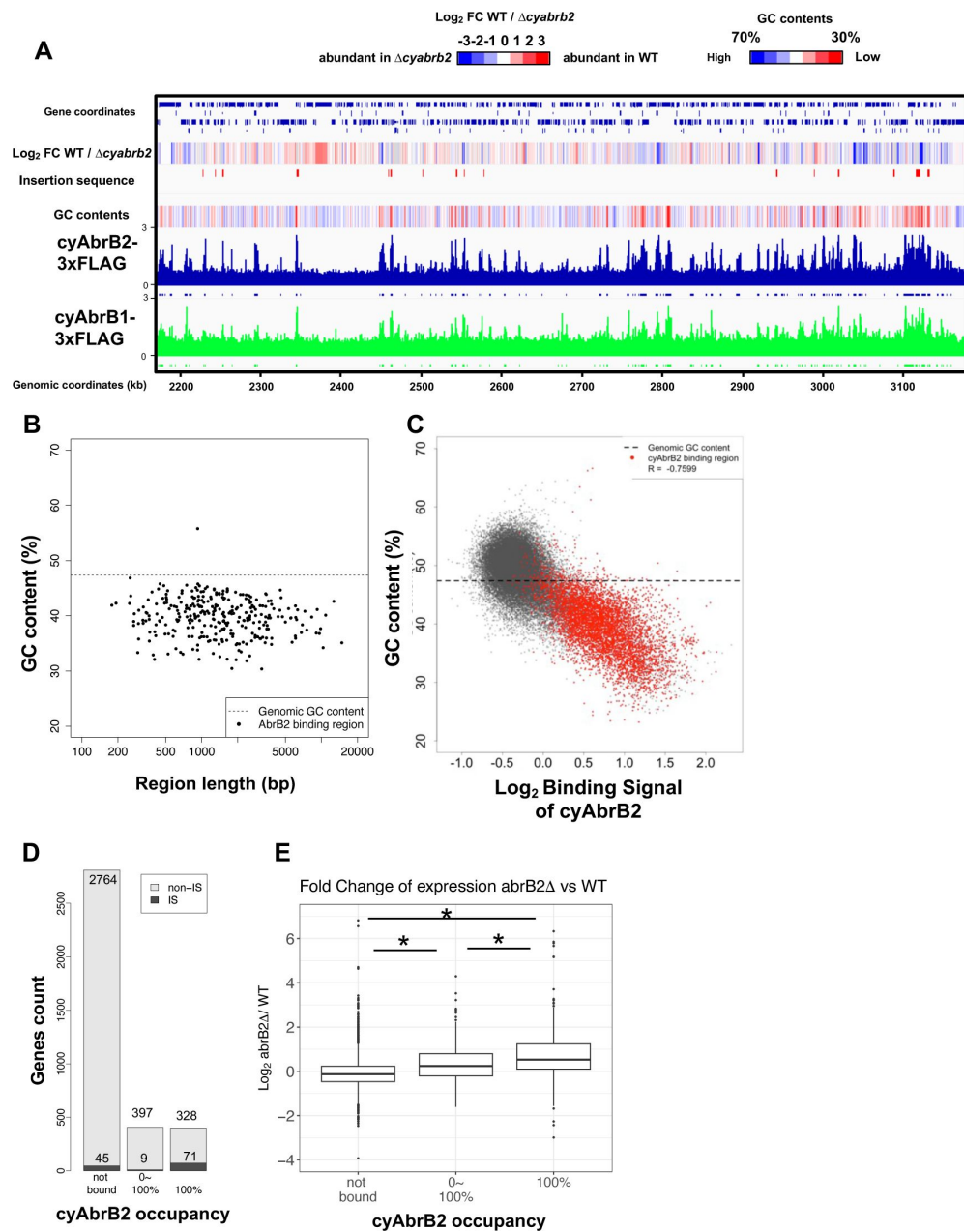


Figure 3

(A) Snapshot of ChIP-seq data for cyAbrB2 and cyAbrB1 under aerobic conditions. The heatmap in the second column indicates expression fold change upon $\Delta cyabr2$ under aerobic conditions. Positive values (colored in red) indicate that the gene expression is higher in wildtype than in $\Delta cyabr2$, and negative values (colored in blue) indicate the opposite. The positions for the insertion elements are marked with red in the third column. The heatmap in the fourth column indicates GC contents. High GC contents are colored in blue and low GC contents are colored in blue. (B) GC contents and region length of cyAbrB2 binding regions (black dots). The horizontal dotted line indicates the genomic average of GC content. (C) Scatter plot of GC content and binding signal of cyAbrB2. The x-axis is the binding signal of cyAbrB2 in each 100 bp region, and the y-axis indicates GC contents within 500 bp windows sliding every 100 base pairs. CyAbrB2 binding regions are marked with red dots. (D) Histogram of genes showing the extent of occupancy (not bound, partially overlapped, or entirely overlapped) by the cyAbrB2 binding region. The gray bars indicate non-IS genes, and the count numbers of the non-IS genes are displayed on the gray bars. The black bars indicate the IS genes, and the count numbers of the IS genes are displayed above the black bars. (E) Boxplot showing fold change in gene expression by $\Delta cyabr2$ under aerobic conditions. Genes are classified according to the extent of occupancy by the cyAbrB2 binding region. Asterisk (*) denotes statistical significance tested by multiple comparisons of the Wilcoxon-rank test.

binding regions located at the transcription start sites (TSSs) and middle of operons, the *pntAB* operon had two binding regions in the middle and downstream of the operon, and the *nrtABCD* operon had one binding region downstream of the operon (**Figure 4C** [↗](#)).

Localization of cyAbrB2 became blurry under the microoxic condition

When cells entered microoxic conditions, the relative ChIP-seq signals in the cyAbrB2 binding regions slightly declined (**Figures 5A** [↗](#) and **5B** [↗](#)). Notably, the total quantities of precipitated DNA by tagged cyAbrB2 did not decrease (**Figure S6A** [↗](#)), and qPCR confirmed that the cyAbrB2 binding signal increased in all positions tested (**Figure 5C** [↗](#)). ChIP-seq data and ChIP-qPCR data indicate that the boundary between cyAbrB2 binding region and cyAbrB2-free region became obscured when the cells entered microoxic conditions due to increased binding of cyAbrB2 to both cyAbrB2 binding and cyAbrB2-free region. The protein amount of cyAbrB2 was not altered on entry to the microoxic condition (**Figure S6B** [↗](#)). The cyAbrB2 binding signal around the transiently upregulated genes became less specific upon entry into microoxic conditions, consistent with the general tendency (**Figure 5B** [↗](#)). The amount of DNA immunoprecipitated by cyAbrB1 was also increased in the microoxic condition, and the protein amount was not increased (**Figures S6C** [↗](#) and **S6D** [↗](#)).

Sigma factors SigE and SigA are excluded from cyAbrB2 binding regions regardless of GC contents

We searched for SigE and SigA binding sites under aerobic and microoxic conditions (**Figures S7A** [↗](#) and **S7B** [↗](#), left and right, respectively). The SigE and SigA peaks identified in this study predominantly covered the previously identified peaks (**Figure S7C** [↗](#)), reproducing the previous study's conclusion ([25](#) [↗](#)); i.e., SigE and the primary sigma factor SigA share localization on the promoters of housekeeping genes, but SigE exclusively binds to the promoters of its dependent genes. SigE and SigA binding peaks were significantly excluded from the cyAbrB2 binding regions (**Figures 6A** [↗](#) and **6B** [↗](#)). SigE preferred the cyAbrB2-free region in the aerobic condition more than SigA did (Odds ratios of SigE and SigA being in the cyAbrB2-free region were 4.88 and 2.74, respectively). CyAbrB2 prefers AT-rich regions, but no correlation was found between the GC content and binding signals of SigE and SigA (**Figures S5A** [↗](#) and **S5B** [↗](#)). Thus, SigA and SigE are excluded from cyAbrB2 binding regions regardless of GC contents.

Dynamics of sigma factors upon exposure to the microoxic condition

When cells entered microoxic conditions, the binding signals of SigA and SigE were changed, although most of their peaks observed under aerobic conditions were present under microoxic conditions (**Figure S7A** [↗](#)). The preference of SigE for the cyAbrB2-free region was alleviated in the microoxic condition (**Figure 6A** [↗](#)). Next, we focused on sigma factor dynamics in transiently upregulated genes. SigE, but not SigA, binds at the TSS of *pntAB* under aerobic and microoxic conditions (**Figure 6C** [↗](#) top). SigE binding summits were not identified at the TSSs of the *hox* and *nifH* operons under aerobic conditions. However, the SigE-specific binding summit appeared at the TSS of *nifH* when cells entered microoxic conditions (**Figure 6C** [↗](#) middle). A bimodal peak of SigE was observed at the TSS of the *hox* operon in a microoxic-specific manner (**Figure 6C** [↗](#) bottom panel). The downstream side of the bimodal SigE peak coincides with the SigA peak and the TSS of TU1715. Another side of the bimodal peak lacked SigA binding and was located at the TSS of the *hox* operon (marked with an arrow in **Figure 6C** [↗](#)), although the peak caller failed to recognize it as a peak. SigE binding without SigA on the promoters of *hox*, *nifH*, and *pntAB* is consistent with their SigE-dependent expression (**Figure 2B** [↗](#)).

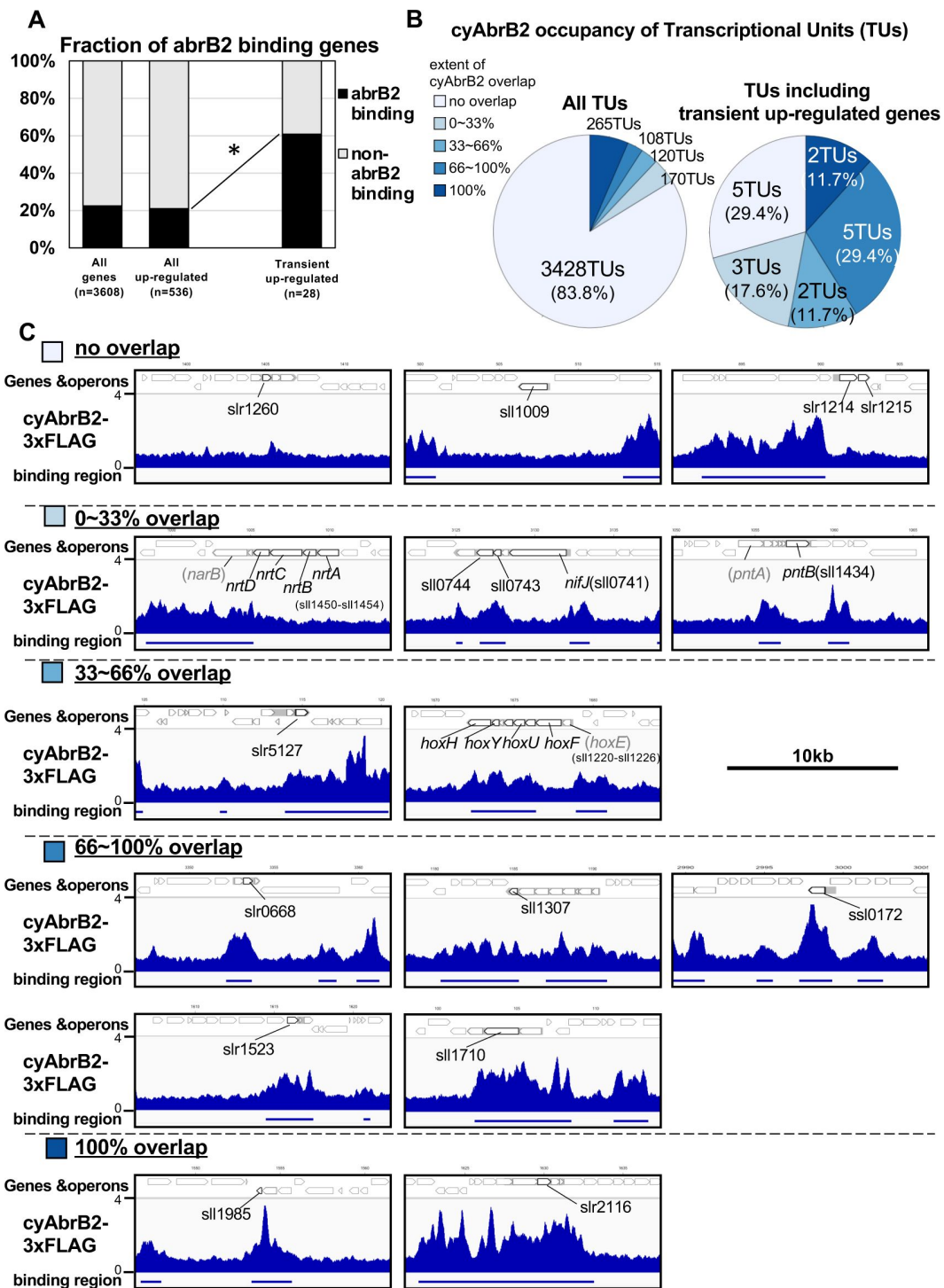


Figure 4

(A) Fraction of genes overlapped or non-overlapped with *cyAbrB2* binding regions at the timepoints of aerobic conditions. Genes are classified according to [Figure S1](#). Asterisk (*) denotes statistically significant enrichment compared with all upregulated genes tested by multiple comparisons of Fisher's exact test. (B) Pie charts of transcriptional units (TUs) classified by extent of overlapping with *cyAbrB2* binding region. The left pie represents all TUs, and the right pie represents only TUs containing the transient upregulated genes. (C) Distribution of *cyAbrB2* in the aerobic condition around transiently upregulated genes. Arrows with bold lines indicate transiently upregulated genes. Shaded arrows indicate operons whose data were obtained from a previous study. The bars below the graph indicate the binding regions of each protein. The black bar at the top of the figure indicates a length of 10 kbp.

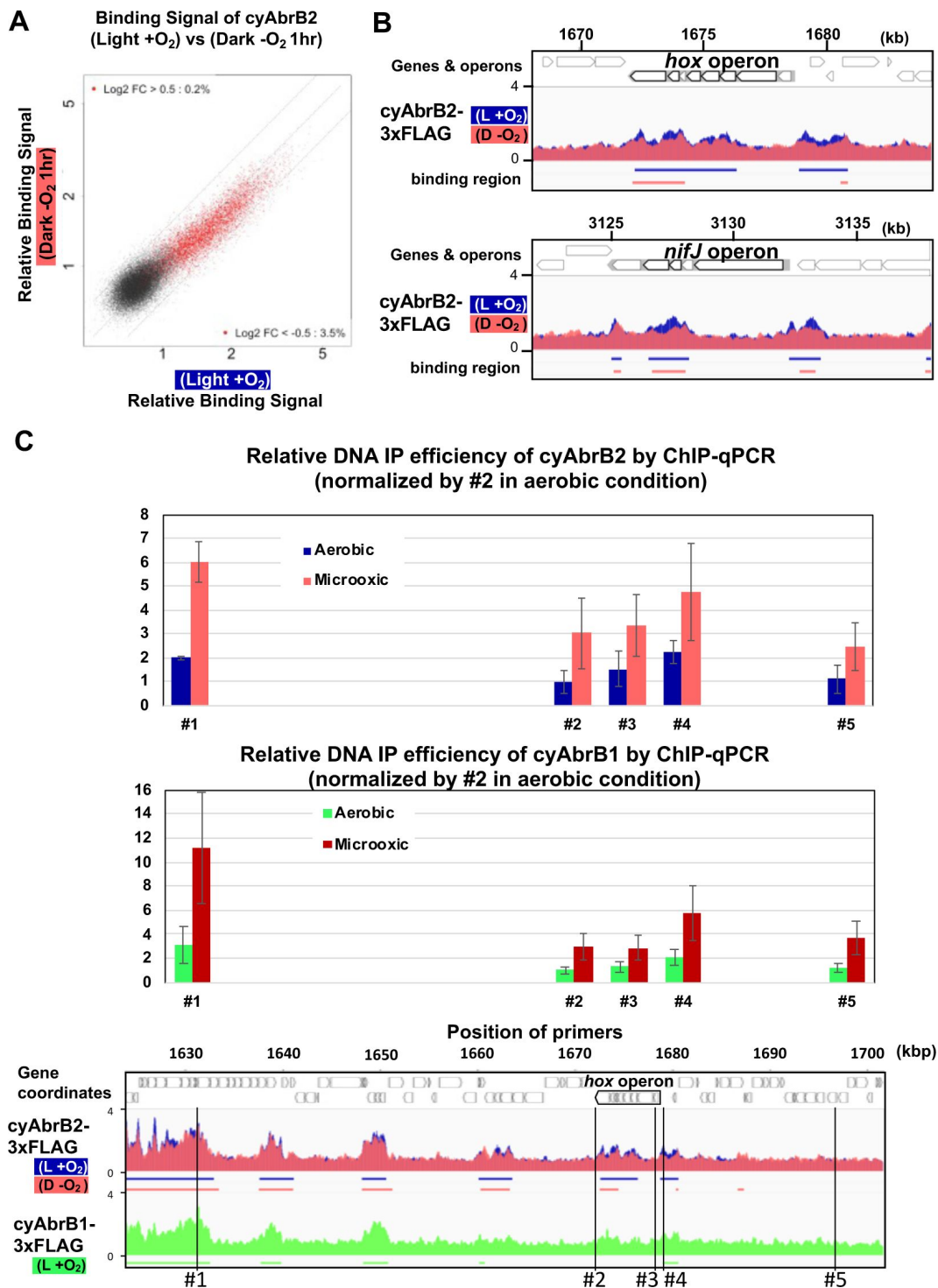


Figure 5

(A) Scatter plot showing changes of the binding signal by 1hr cultivation in the microoxic condition. The binding signal of each 100bp window is plotted. Red dots are cyAbrB2 binding regions in either aerobic or microoxic conditions. The dotted lines indicate Log2 fold enrichment of 0.5, 0, and -0.5 between aerobic and microoxic conditions. (B) Distribution of cyAbrB2 around *hox* operon and *nifJ* operon. ChIP-seq data in aerobic (L + O₂) and dark microoxic (D - O₂) conditions are overlaid. The bars below the graph indicate the binding regions of each protein. (C) Quantification for IP efficiency of cyAbrB2 (top) and cyAbrB1 (middle) by qPCR in the aerobic and microoxic conditions. The position of primers and ChIP-seq data of cyAbrB2 are shown at the bottom. Scores are normalized by the IP% at position #2 in the aerobic condition.

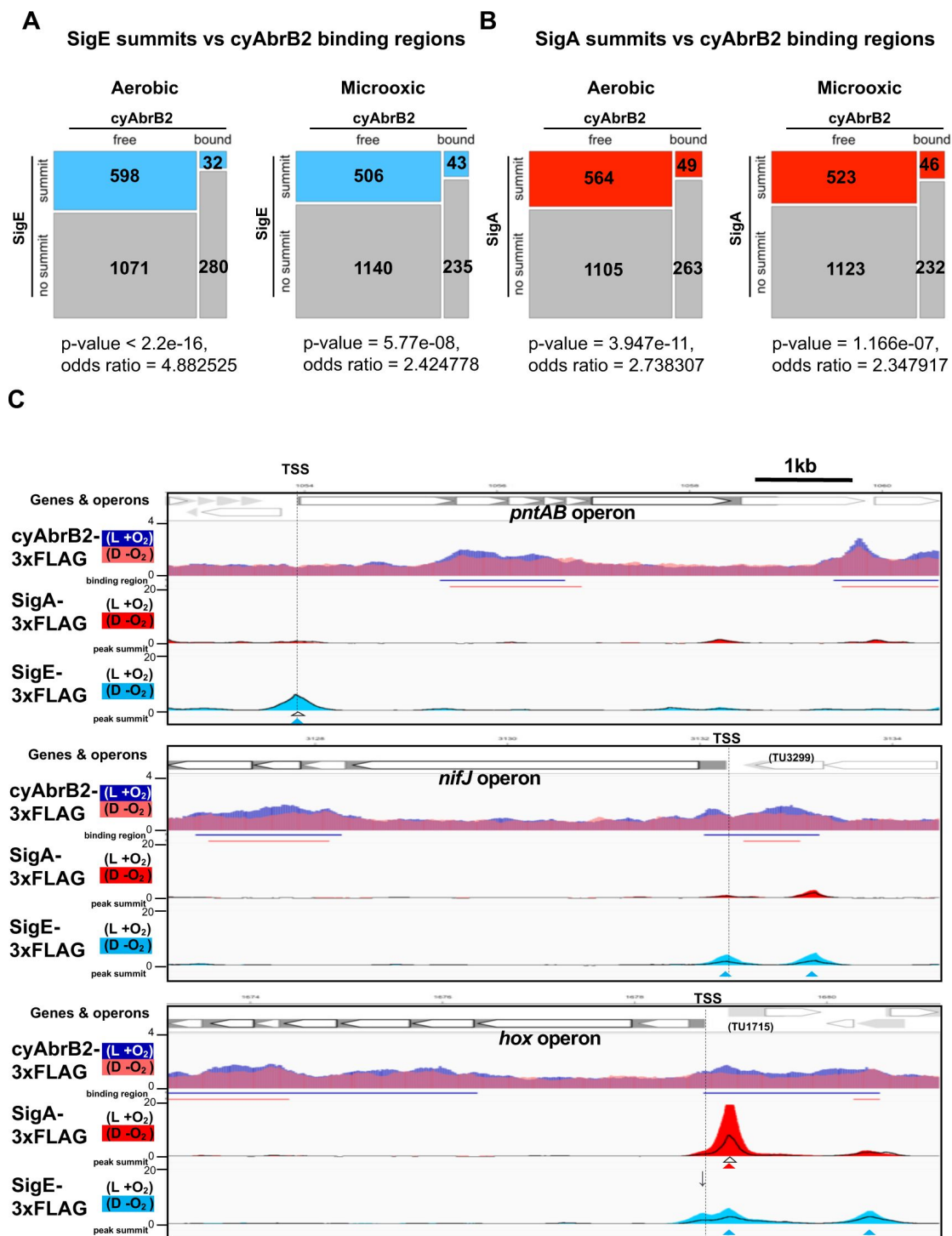


Figure 6

(A, B) Anti-co-occurrence of cyAbrB2 binding regions and sigma factors. Mosaic plots of cyAbrB2 binding regions and SigE peaks (A) or SigA binding peaks (B) are shown. Odds and p-values were calculated by Fisher's exact test. (C) Snapshots of ChIP-seq data for CyAbrB2, SigE, and SigA at the *nifJ* region (top) and *hox* region (bottom). ChIP-seq data for cyAbrB2 under aerobic and microoxic conditions are colored blue and pink, respectively. ChIP-seq data for SigE and SigA are shown in solid lines (aerobic conditions) and the area charts (microoxic conditions). The positions of TSSs were obtained from a previous study (Kopf et al., 2014) and indicated by vertical dotted lines. Open triangles indicate peak summits under aerobic conditions, and solid triangles indicate peak summits under microoxic conditions.

Chromatin conformation around *hox* operon and *nifj* operon

We have shown that cyAbrB2 broadly binds to AT-rich genomic regions, including insertion element sequences, and represses expression (**Figure 3**). This is functionally similar to the NAPs (14), which makes us hypothesize that cyAbrB2 modulates chromosomal conformation. Therefore, we conducted the chromatin conformation capture (3C) assay against wildtype and *cyabrb2Δ* strains at aerobic and microoxic conditions. qPCR was performed with unidirectional primer sets, where the genomic fragment containing *hox* operon and *nif* operon (hereinafter *hox* fragment and *nifj* fragment, respectively) were used as bait (**Figure 7**).

First, focusing on the aerobic condition of wildtype (**Figure 7B**, solid line), the *hox* fragment interacted with its proximal downstream loci (loci (f) to (g)) and proximal upstream locus (locus (j)). The *hox* fragment also interacts with the distal downstream locus (locus (c)). Meanwhile, the *nifj* fragment shows high interaction frequency with proximal upstream and downstream loci (**Figure 7G**, loci (i') and (j')), and a distal downstream locus (locus (g')) showed higher interaction frequency with *nifj* fragment than proximal locus (h') did. The upstream regions of *nifj* (loci (l') to (n') and (p')) showed comparable frequency with locus (g').

The chromatin conformation is changed in *cyabrb2Δ* in some loci

Then we compared the chromatin conformation of wildtype and *cyabrb2Δ*. Although overall shapes of graphs did not differ, some differences were observed in wildtype and *cyabrb2Δ* (**Figures 7B** and **7G**); interaction of locus (c) with *hox* region were slightly lower in *cyabrb2Δ* and interaction of loci (f') and (g') with *nifj* region were different in wildtype and *cyabrb2Δ*. Note that the interaction scores exhibit considerable variability and we could not detect statistical significance at those loci.

Changes of chromatin conformation upon microoxic condition

When the cells entered the microoxic condition, proximal loci interacted more frequently (**Figure 7D**; loci (f)-(h) and **Figure 7I** loci (j') and (k')). This tendency was more apparent in *cyabrb2Δ* (**Figures 7E** and **7J**). Furthermore, the interaction of *nifj* upstream loci (l') – (n') increased in the microoxic condition in *cyabrb2Δ* but not wildtype (**Figures 7I** and **7J**). The locus (c) and locus (j) interacted less frequently with *hox* fragment upon entry to the microoxic condition in the wildtype. While the interaction scores exhibit considerable variability, the individual data over time demonstrate declining trends of the wildtype at locus (c) and (j) (**Figure S8**). In *Δcyabrb2*, by contrast, the interaction frequency of loci (c) and (j) was unchanged in the aerobic and microoxic conditions (**Figure 7E**). The interaction frequency of locus (c) in *Δcyabrb2* was as low as that in the microoxic condition of wildtype, while that of locus (j) in *Δcyabrb2* was as high as that in the aerobic condition of wildtype (**Figures 7B** and **7C**). In summary, 3C analysis demonstrated cyAbrB2-dependent and independent dynamics of chromosomal conformation around the *hox* and *nifj* operon in response to the microoxic condition (**Figure 8**).

Discussion

Physiological significance of transient upregulation of *hox* and *nifj* operons

As the transcriptional change can alter the metabolic flow, the transcriptional upregulation of fermentative genes in response to the microoxic condition is expected to be adaptive for energy acquisition and the maintenance of redox balance. Our time-course transcriptome showed upregulation of several genes involved in catabolism upon exposure to the microoxic condition. The transient upregulation of *hox* and *nifj* operons is distinctive among them (**Figure 1D**).

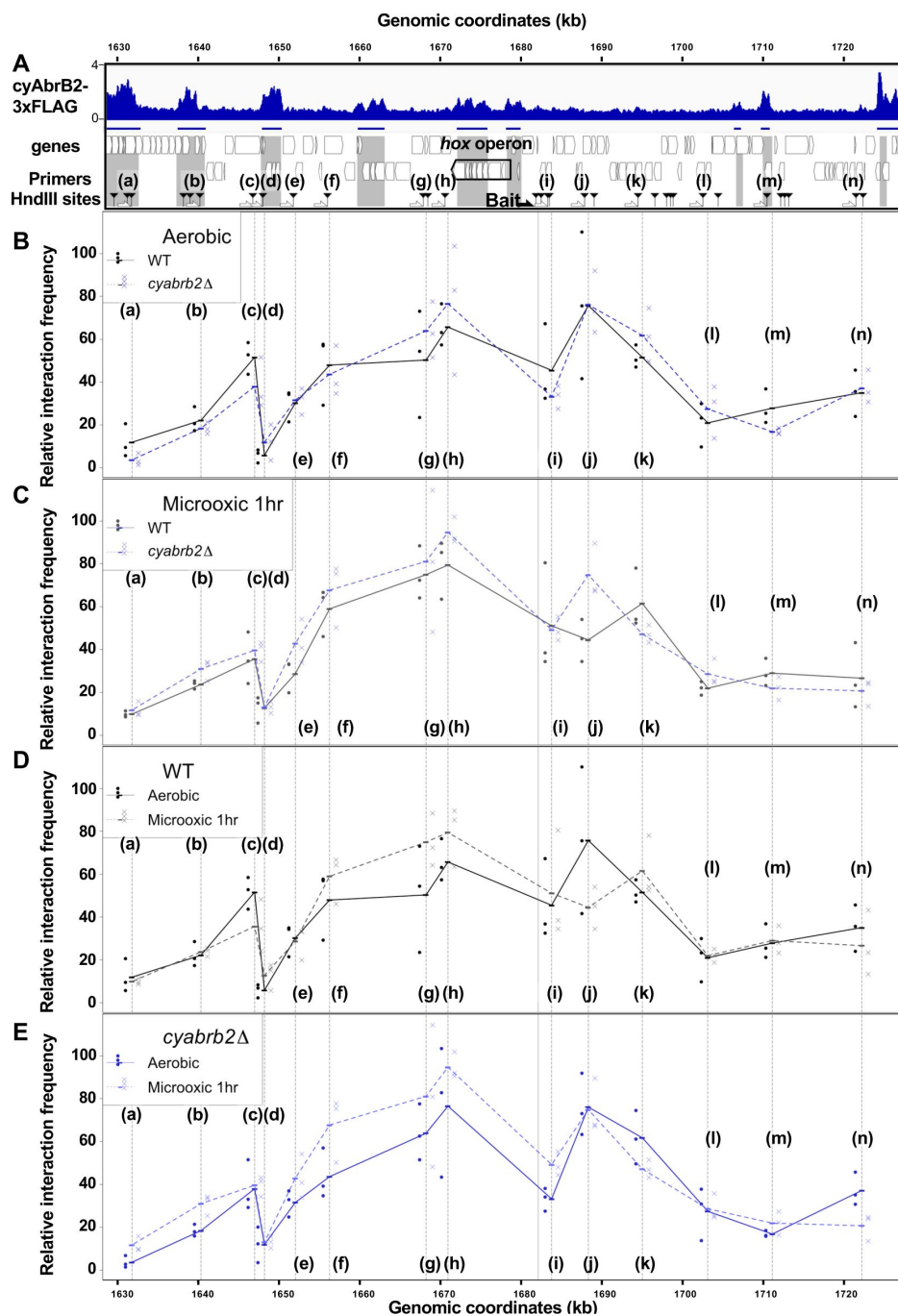


Figure 7

(A) and (F) Schematic diagram of 3C analysis around *hox* operon (A) and *nifH* operon (F). In the panels (A) and (F), the black horizontal arrow shows the location of the bait primer, and white horizontal arrows ((a) to (n) in *hox* operon (A) and (a') to (t') in *nifH* operon (F)) indicate loci where the interaction frequency with bait were assayed. Vertical black arrowheads indicate the position of HindIII sites. ChIP-seq data of cyAbrB2 in the aerobic condition is displayed in the bottom, and cyAbrB2 binding regions are marked with shade. (B - E) The line plot showing the interaction frequency of each locus with *hox* fragment. Two of data sets are presented; (B) wildtype vs $\Delta cyabrb2$ in aerobic condition, (C) wildtype vs $\Delta cyabrb2$ in 1hr of microoxic condition, (E) wildtype in aerobic vs 1hr of microoxic condition, and (E) $\Delta cyabrb2$ in aerobic vs 1hr of microoxic condition are compared. (G - J) The line plot showing the interaction frequency of each locus with *nifH* fragment. Two data sets are selected and presented; (G) wildtype vs $\Delta cyabrb2$ in aerobic condition, (H) wildtype vs $\Delta cyabrb2$ in 1hr of microoxic condition, (I) wildtype in aerobic vs 1hr of microoxic condition, and (J) $\Delta cyabrb2$ in aerobic vs 1hr of microoxic conditions are compared. The line plots indicate the average interaction frequency over the random ligation (n=3). Individual data are plotted as dots.

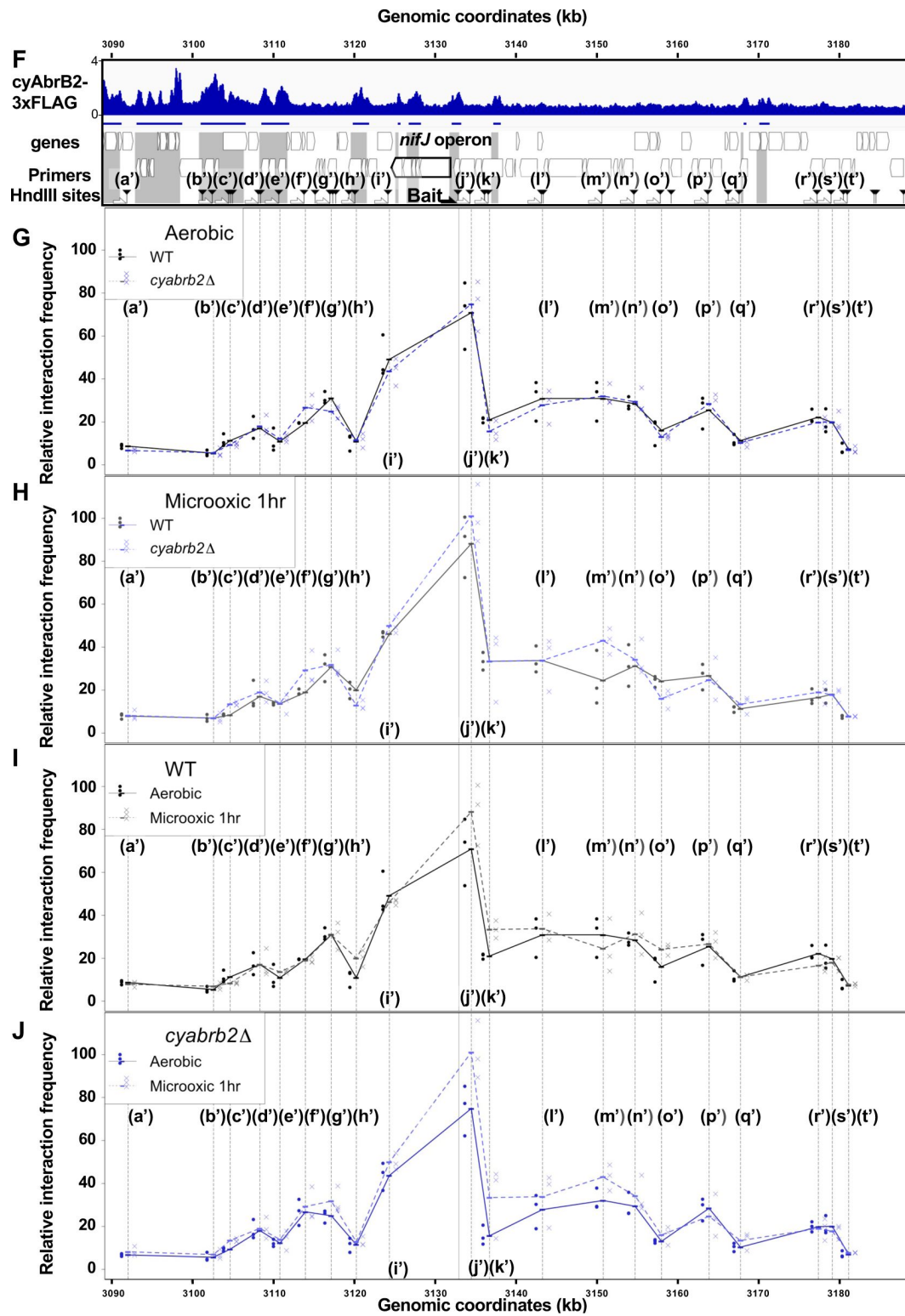


Figure 7 (continued)

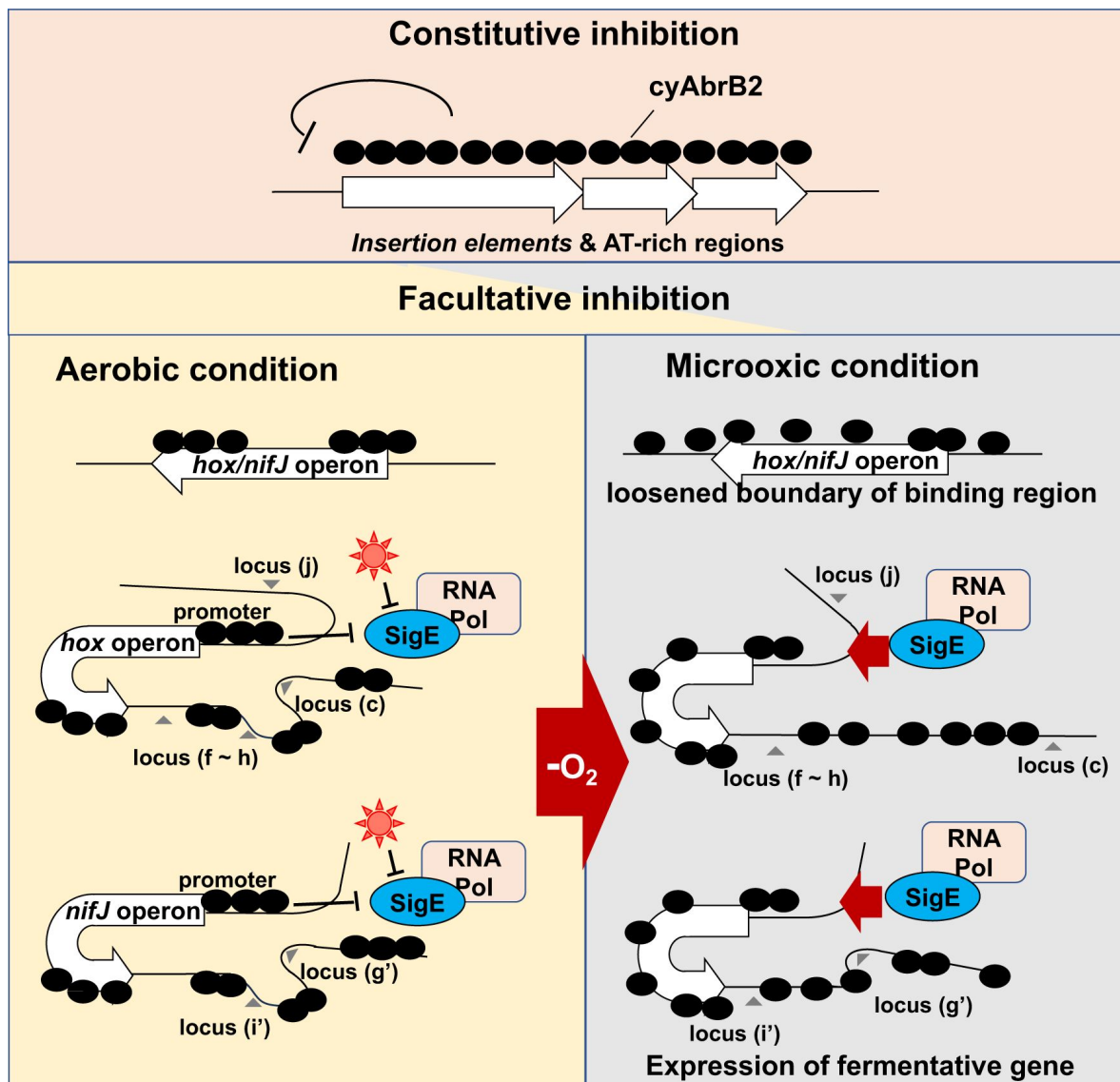


Figure 8

Schematic diagram of the dynamics of transcription factors governing fermentative gene expression.

One reason for transient upregulation is probably the resource constraints of inorganic cofactors. Hydrogenase and PFOR (the product of *nifH* gene) have iron-sulfur clusters, and hydrogenase requires nickel for its activity (32, 33). Overexpression of the *hox* operon should be futile under physiological conditions without an adequate nickel supply (34).

Another significance for transient upregulation may be the reusability of fermentative products. Hydrogen, lactate, and dicarboxylic acids can be reused as the source of reducing power when cells return to aerobic conditions (12, 35, 36). The substrate proton is abundant, but hydrogen is diffusive and difficult to store. Therefore, hydrogenase may favor fermentation initiation, and the reductive branch of TCA-producing dicarboxylic acids may become active subsequently. In fact, *citH/mdh* (sll0891) encoding a key enzyme of the reductive branch of TCA was classified as continuously upregulated genes in this study (Figures 1C and 1D).

Mechanisms for transient expression mediated by SigE and cyAbrB2

SigE and cyAbrB2 can independently contribute to the transient transcriptional upregulation. This is evident as the single mutants, $\Delta sigE$ or $\Delta cyabrB2$, maintained transient expression of *hoxF* and *nifH* (Figure S2). We first discuss cyAbrB2 as the potential NAPs, and then the mechanism of transient upregulation mediated by cyAbrB2 and SigE will be discussed.

cyAbrB2 is a novel nucleoid-associated protein of cyanobacteria

We have shown that cyAbrB2 broadly binds to AT-rich genomic regions, including IS elements (Figure 3). This is functionally similar to the histone-like nucleoid protein H-NS family; including H-NS in *Enterobacteriaceae* (37, 38), and Lsr2 in *Mycobacteria* (39). Like H-NS and Lsr2, cyAbrB2 may defend against exogenous DNA elements, which often have different GC content. Interestingly, Lsr2 controls genes responding to hypoxia, showing a functional analogy with cyAbrB2 (40).

The biochemistry of cyAbrB2 will shed light on the regulation of chromatin conformation in the future

H-NS proteins often cause bound DNA to bend, stiffen, and/or bridge (14). DNA-bound cyAbrB2 is expected to oligomerize, based on the long tract of cyAbrB2 binding region in our ChIP-seq data. However, no biochemical data mentioned the DNA deforming function or oligomerization of cyAbrB2 in the previous studies, and preference for AT-rich DNA is not fully demonstrated in vitro (19, 27, 41). Moreover, our 3C data did not support bridging at least in *hox* region and *nifH* region, as the high interaction locus and cyAbrB2 binding region did not seem to correlate (Figure 7). Therefore, direct observation of the DNA-cyAbrB2 complex by atomic force microscopy is the solution in the future.

Not only DNA structural change but also the effect of the post-translational modification can be investigated by biochemistry. The previous studies report that cyAbrB2 is subject to phosphorylation and glutathionylation (42, 43), and pH and redox state alters cyAbrB1's affinity to DNA (41). Those modifications might respond to environmental changes and be involved in transient expression. Overall, the biochemistry integrating assay conditions (PTM, buffer condition, and cooperation with cyAbrB1) and output (DNA binding, oligomerization, and DNA structure) will deepen the understanding of cyAbrB2 as cyanobacterial NAPs.

Cooperative and antagonistic function of cyAbrB1 and cyAbrB2

CyAbrB1, the homolog of cyAbrB2, may cooperatively work, as cyAbrB1 directly interacts with cyAbrB2 (29), their distribution is similar, and they partially share their target genes for suppression (Figures 3A, S5C and S5D). The possibility of cooperation would be examined

by the electrophoretic mobility shift assay of cyAbrB1 and cyAbrB2 as a complex. Despite their similar repressive function, cyAbrB1 and cyAbrB2 regulate *hox* expression in opposite directions, and their mechanism remains elusive. The stoichiometry of cyAbrB1 and cyAbrB2 bound to DNA fluctuates in response to the environmental changes (28), but there was no difference in the behavior of cyAbrB1 and cyAbrB2 around the *hox* region on entry to the microoxic condition.

Localization pattern and function of cyAbrB2

Herein, we classified three types of binding patterns for cyAbrB2. The first is that cyAbrB2 binds a long DNA tract covering the entire gene or operon, represented by the insertion sequence elements. CyAbrB2 suppresses expression in this pattern (Figure 3E). In the second pattern, cyAbrB2 binds on promoter regions, such as *hox* operon and *niff*. The binding on those promoters fluctuates in response to environmental changes, thus regulating expression. This pattern also applies to the promoter of *sbtA* ($\text{Na}^+/\text{HCO}_3^-$ symporter), where cyAbrB2 is bound in a CO_2 concentration-dependent manner (28). The last one is cyAbrB2 binding in the middle or downstream of operons. The middle of *hox*, *pntAB*, and *niff* operons and the downstream of *nrt* operon are the cases (Figure 4C). Our data show that genes in the same operon separated by the cyAbrB2 binding region behave differently. In particular, *pntB* is classified as the transiently upregulated gene, while *pntA* is not, despite being in the same operon. This might be explained by the recent study which reported that cyAbrB2 affects the stability of mRNA transcribed from its binding gene (41). The cyAbrB2-mediated stability of mRNA may also account for the decrease in transcript from transient upregulated genes at 4 hours of cultivation. Hereafter, we will focus on the mechanism of the second pattern, regulation by cyAbrB2 on the promoter.

Insight into the regulation of *hox* and *niff* operon by cyAbrB2

Genome-wide analysis indicates that the cyAbrB2-bound region blocks SigE and SigA (Figures 6A and 6B). This is presumably because sigma factors recognize the promoter as a large complex of RNA polymerase. CyAbrB2 binds to the *hox* and *niff* promoter region and may inhibit access to RNA polymerase complex under aerobic conditions. When cells entered microoxic conditions, the boundaries of the cyAbrB2 binding region and cyAbrB2-free region became obscure (Figure 5), and SigE binding peaks on those promoters became prominent (Figure 6C). Notably, cyAbrB2 ChIP efficiency at the *hox* promoter is higher in the microoxic condition than in the aerobic condition (Figure 5). Hence, while the exclusion by cyAbrB2 occupancy on promoter inhibits containing RNA polymerase in the aerobic condition, it is also plausible that chromosomal conformation change governed by cyAbrB2 provides SigE-containing RNA polymerase with access to the promoter region (Figure 8). Our 3C result demonstrated that cyAbrB2 influences the chromosomal conformation of *hox* and *niff* region to some extent (Figure 7).

A recent study demonstrated that manipulating the expression of topoisomerase, which influences chromosomal conformational change through supercoiling, affects transcriptional properties in cyanobacteria (15). Moreover, Song et al. pointed out that DNA looping may inhibit transcription in cyanobacteria (41) because artificial DNA looping by the LacI repressor of *E. coli* can repress cyanobacterial transcription (44). Thus, we infer conformation change detected by the present 3C experiment regulates expression of *hox* operon.

Generality for chromosomal conformation in cyanobacteria

Our 3C analysis revealed that local chromosomal conformation changes upon entry to the microoxic conditions (Figure 8). As cyAbrB2 occupies about 15% of the entire genome and globally regulates gene expression, cyAbrB2 likely governs the whole chromosomal conformation. Furthermore, the conformational changes by deletion of cyAbrB2 were limited, suggesting there are potential NAPs in cyanobacteria yet to be characterized. It is speculated that conformational change of the entire chromosome occurs to deal with many environmental stresses.

The sigE-mediated mechanism for the transient expression

One possible SigE-mediated mechanism for transient expression is the post-transcriptional activation and degradation of SigE in the dark; i.e., SigE is sequestered by anti-sigma factor under light conditions and released under dark (45 [↗](#)), enabling acute transcription of *hox* operon and *nifH*. Transcripts of *sigE* were continuously downregulated in our time-course transcriptome, while *sigB* (sll0306) and *sigC* (sll0184) were classified as continuous upregulated genes (Supplemental Table S1 [↗](#)). It is possible that upregulated SigB and SigC outcompete SigE in prolonged incubation under microoxic conditions. Finally, SigE is degraded under dark within 24 hours (46 [↗](#)).

Another reason for the microoxic specific expression may exist in the sequence of the *hox* promoter. We previously determined the consensus sequence of -10 element for SigE regulon in the aerobic condition as “TANNNT,” where N is rich in cytosine (25 [↗](#)). The -10 sequence of the *hox* promoter “TAACA” (23 [↗](#)) deviates from the consensus, and no hexamer precisely fitting the consensus is found in the *nifH* promoter. This deviation can inhibit SigE from binding during aerobic conditions, aside from cyAbrB2-mediated inhibition. Under the microoxic condition, transcription factors LexA (23 [↗](#)) and Rre34 (16 [↗](#)) may aid SigE binding to the promoter of *hox* and *nifH*, respectively.

Moreover, SigE seems susceptible to the blocking from cyAbrB2 during the aerobic condition compared with SigA. This is supported by the odds ratio of SigE being in the cyAbrB2-free region was higher than that of SigA in the aerobic condition (Figures 6A [↗](#) and 6B [↗](#)). The higher exclusion pressure of cyAbrB2 on SigE may contribute to sharpening the transcriptional response of *hox* and *nifH* on entry to microoxic conditions. Overall, multiple environmental signals are integrated into the *hox* and *nifH* promoter through the cyAbrB2 and SigE dynamics.

Materials and Methods

Bacterial strains and plasmids

The glucose-tolerant strain of *Synechocystis* sp. PCC 6803 (47 [↗](#)) was used as a wildtype strain in this study. The *sigE* (sll1689)-disrupted strain (G50), SigE FLAG-tagged strain, and SigA FLAG-tagged strain were constructed in a previous study (17 [↗](#), 25 [↗](#)). Disruption and epitope-tagging of *cyabrb1* (sll0359) and *cyabrb2* (sll0822) were performed by homologous double recombination between the genome and PCR fragment (47 [↗](#)). The resulting transformants were selected using three passages on BG-11 plates containing 5 μ g/mL kanamycin. Genomic PCR was used to confirm the insertion of epitope tag fragments and gene disruption (Figure S9 [↗](#)). Supplemental Tables S2 [↗](#), S3 [↗](#), and S4 [↗](#) contain the cyanobacterial strains, oligonucleotides, and plasmids used in this study.

Aerobic and microoxic culture conditions

For aerobic conditions, cells were harvested after 24 hours cultivation in HEPES-buffered BG-11₀ medium (48 [↗](#)), which was buffered with 20 mM HEPES-KOH (pH 7.8) containing 5 mM NH₄Cl under continuous exposure to white light (40 μ mol m⁻²s⁻¹) and bubbled with air containing 1% CO₂ (final OD₇₃₀ = 1.4–1.8). For the dark microoxic culture, the aerobic culture cell was concentrated to an OD₇₃₀ of 20 with the centrifuge and resuspended in the culture medium. The concentrated cultures were poured into vials, bubbled with N₂ gas, and sealed. The sealed vials were shaded and shaken at 30°C for the described times.

Antibodies and immunoblotting

Sample preparation for immunoblotting was performed as previously described (25 [↗](#)), and FLAG-tagged proteins were detected by alkaline-phosphatase-conjugated anti-FLAG IgG (A9469, Sigma Aldrich, St. Louis, MO) and 1-Step NBT/BCIP substrate solution (Thermo Fisher Scientific, Waltham, MA).

RNA isolation

Total RNA was isolated with ISOGEN (Nippon gene, Tokyo, Japan) following the manufacturer's instructions and stored at -80°C until use. The extracted RNA was treated with TURBO DNase (Thermo Fisher Scientific) for 1 hour at 37°C to remove any genomic DNA contamination. We confirmed that the A260/A280 of the extracted RNA was >1.9 by NanoDrop Lite (Thermo Fisher Scientific). We prepared triplicates for each timepoint for the RNA-seq library. RT-qPCR was performed as described elsewhere (46 [↗](#)).

ChIP assay

Two biological replicates were used for each ChIP-seq experiment, and one untagged control ChIP was performed. ChIP and qPCR analyses were performed using the modified version of a previous method (25 [↗](#)). FLAG-tagged proteins were immunoprecipitated with FLAG-M2 antibody (F1804 Sigma-Aldrich) conjugated to protein G dynabeads (Thermo Fisher Scientific).

Library preparation and next-generation sequencing

For the ChIP-seq library, input and immunoprecipitated DNA were prepared into multiplexed libraries using NEBNext Ultra II DNA Library Prep Kit for Illumina (New England Biolabs, Ipswich, MA). For the RNA-seq library, isolated RNA samples were depleted of ribosomal RNA with Illumina Ribo-Zero Plus rRNA Depletion Kit (Illumina, San Diego, CA) and processed into a cDNA library for Illumina with the NEBNext Ultra II Directional RNA Library Prep Kit for Illumina (New England Biolabs). Dual-index primers were conjugated with NEBNext Multiplex Oligos for Illumina (Set1, New England Biolabs). We pooled all libraries, and the multiplexed libraries were dispatched to Macrogen Japan Inc. and subjected to paired-end sequencing with HiSeqX. Adapter trimming and quality filtering of raw sequence reads were conducted with fastp (ver. 0.21.0) (49 [↗](#)) under default conditions. The paired-end sequences were mapped onto the *Synechocystis* genome (ASM972v1) using Bowtie2 (50 [↗](#)) (ver. 2.4.5 paired-end). Supplementary Table S5 contains the read counts that passed via fastp quality control and were mapped by Bowtie2.

RNA-seq analysis

Mapped reads were counted by HT-seq count (ver. 2.0.2) (51 [↗](#)) for the GFF file of ASM972v1, with the reverse-strandedness option. EdgeR package (ver. 3.40.1) (52 [↗](#)) was used to perform the differential expression analysis. Fold changes in expression and FDR were used for gene classification. Supplemental Table S6-S8 contains fold change in gene expression calculated by edgeR.

Genome-wide analyses

Peaks were called using the MACS3 program (ver. 3.0.0b1) (53 [↗](#)). For paired-end reads for SigE, SigA, and untagged control ChIP, narrow peaks were called with $<1e-20$ of the q-value cut-off and “--call-summits” options. The peak summits from two replicates and the untagged control were merged if summits were located within 40 bp of each other. Peak summits identified in both replicates but not in the control, were considered for further analysis. The midpoint of the peak summits for the two merged replicates was further analyzed.

Broad peak calling methods were applied to paired-end reads for cyAbrB2, cyAbrB1, and untagged control ChIP using the “-broad” option, with a q-value cut-off of < 0.05 and a q-value broad cut-off of < 0.05. The intersection of broad peaks from two replicates, excluding those called by the control, was used in subsequent analyses.

The positions of the TSS, including internal start sites, were obtained as reported by (31). The read count, merging, and intersection of the binding region were calculated using BEDTools (ver. 2.30.0) (54). Supplemental Tables S9–S15 contain SigA and SigE peaks and the broad binding regions of cyAbrB2 and cyAbrB1, respectively.

Binding signals in every 100 bp bin for scatter plots were calculated as (IP read counts within 100 bp window) / (input read counts within 100 bp window) * (total input read counts/total IP read counts). GC contents were calculated within 500 bp in 100 bp sliding windows by seqkit (ver. 2.3.0) (55).

Genome extraction, digestion, and ligation for 3C assay

A 3C assay was conducted based on the previous prokaryotic Hi-C experiment (56, 57), with certain steps modified. To begin, *Synechocystis* were fixed with 2.5% formaldehyde for 15 minutes at room temperature. Fixation was terminated by adding a final concentration of 0.5M of glycine, and cells were stored at –80°C until use. Fixed cells were disrupted using glass beads and shake master NEO (Bio Medical Science, Tokyo, Japan), following the previous study’s instructions for preparing cell lysate for ChIP. The lysates were incubated with buffer containing 1mM Tris-HCl (pH7.5), 0.1mM EDTA, and 0.5% SDS for 10min at room temperature, and 1% Triton X-100 quenched SDS. Genomes in the cell lysate were digested by 600U/mL of HindIII (Takara Bio, Shiga, Japan) for 4 hours at 37°C, and RNA in the lysate was simultaneously removed by 50 µg/mL of RNaseA (Nippon genetics, Tokyo, Japan). The digestion was terminated by adding 1% SDS and 22 mM EDTA. The fill-in reaction and biotin labeling steps were omitted from the procedure. The digested genomes were diluted by ligation buffer containing 1% triton-X100 to the final concentration of approximately 1µg/mL and incubated for 10 min at room temperature. Ligation was performed with 2 U/mL of T4 DNA ligase (Nippon Gene) overnight at 16°C. Crosslinking was reversed under 65°C for 4 hours in the presence of 2.5 mg/mL of proteinase K (Kanto Chemical, Tokyo, Japan), and DNA was purified with the phenol-chloroform method and ethanol precipitation method.

Preparation of calibration samples for 3C qPCR

Based on a previous study, calibration samples for possible ligated pairs were prepared in parallel with 3C ligation (58). In brief, the purified genome of *Synechocystis* was digested by HindIII, and DNA was purified with the phenol-chloroform and ethanol precipitation. Purified DNA was dissolved into the ligation buffer at a concentration of about 600ng/µL and ligated with 2 U/mL of T4 DNA ligase at 16°C overnight.

Quantification of crosslinking frequency for 3C assay

Before the real-time PCR assay, we confirmed that each primer set amplified single bands in a ligation-dependent manner by GoTaq Hot Start Green Master Mix (Promega, Madison, WI) (Figure S10). Real-time PCR was performed with StepOnePlus (Applied Biosystems, Foster City, CA) and Fast SYBR Green Master Mix (Thermo Fisher Scientific) according to the manufacturer’s instructions. Interaction frequency was calculated by $\Delta\Delta C_t$ method using dilution series of calibration samples described above. We confirmed each primer set amplified DNA fragment with a unique T_m value. The amount of the bait fragment containing *hox* operon were quantified and used as an internal control. Supplemental Table S4 contains the list of primers used in the 3C

quantification. Interaction frequency for each primer position was calculated as the relative abundance of ligated fragments against the calibration samples and normalized among samples by internal control.

Statistical analysis

Statistical analyses were performed with R version 4.2.2 ([59](#)). The “fisher.test” function was used for Fisher’s exact test, and p-values < 0.05 were denoted as asterisks in the figure. Multiple comparisons of Fisher’s exact test were conducted using “fisher.Multcomp” function in the RVAideMemoire package ([60](#)), where p-values were adjusted by the “fdr” method and FDRs < 0.05 are shown in the figures. Multiple comparisons of the Wilcoxon-rank test were conducted by “pairwise.wilcox.test,” and p-values were adjusted by the “fdr” method. Adjusted p-values < 0.05 are shown in the figure. The correlation coefficient was calculated with the “cor” function. Gene set enrichment analysis (GSEA) was performed by culsterPlofiler package ([61](#)) in R with p-value cut-off of 0.05. The enriched pathways detected by GSEA are listed in Supplemental Table S16.

Manuscript writing

We employed GPT-3.5 to refine the expression of the manuscripts.

Accession Numbers

Raw ChIP sequencing and RNA sequencing reads were deposited in the Sequence Read Archive (accession ID: PRJNA956842).

Declaration of interest

The authors declare no competing interests.

Acknowledgements

This study was supported by the following grants to T.O.: Grant-in-Aid for Scientific Research (B) (grant no. 20H02905), JST-ALCA of the Japan Science and Technology Agency (grant number JPMJAL1306), the Asahi Glass Foundation, and JST-GteX (grant number JPMJGX23B0). We thank Dr. Kohki Yoshimoto for providing laboratory instruments and Ms. Kaori Iwazumi for the support of bacterial culture and the medium preparation.

Author contributions

R.K. and T.O. designed the study; R.K. conducted the experiment; R.K. analyzed the data; R.K. and T.O. wrote the paper.

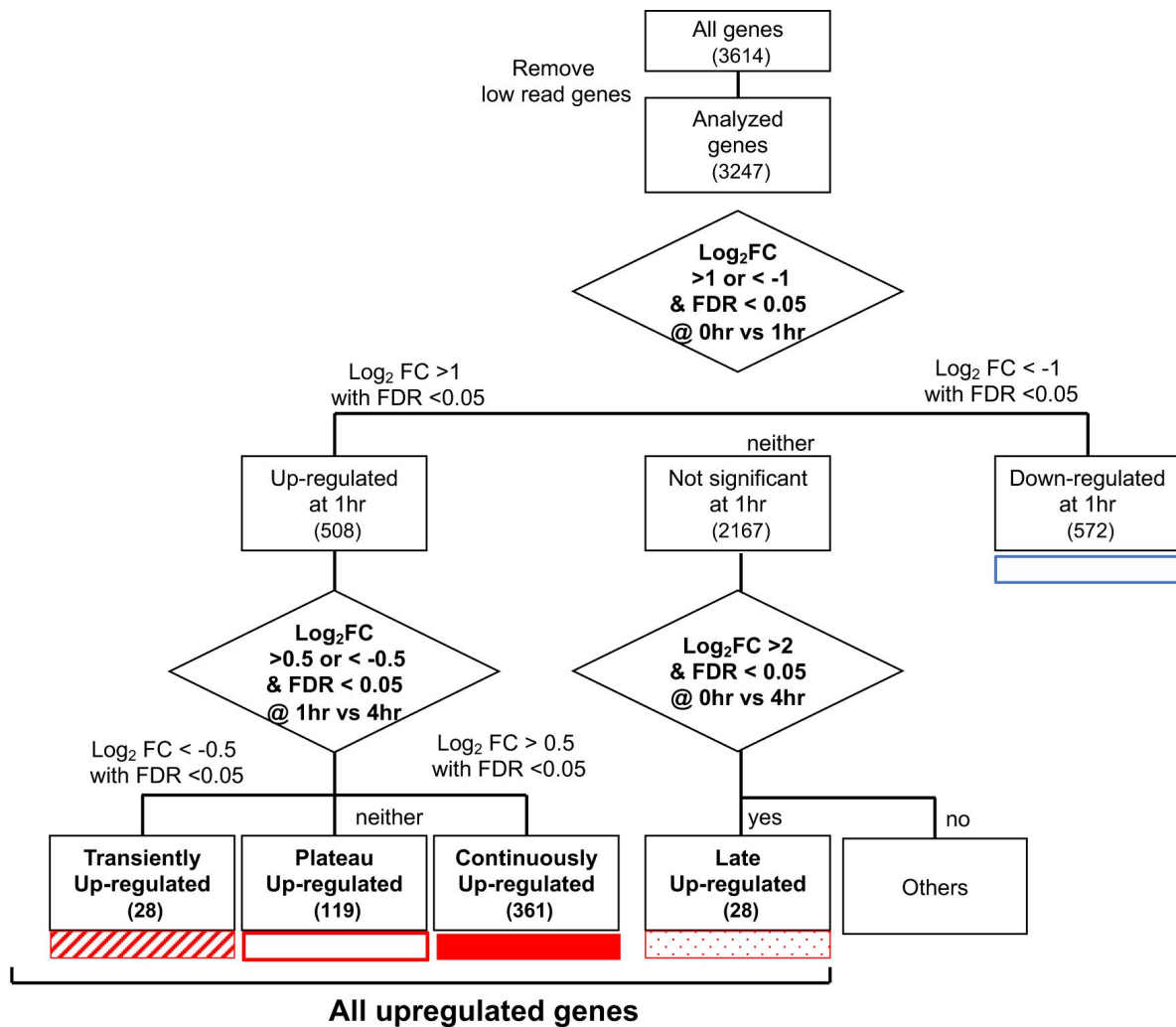


Figure S1

Schematic diagram showing the classification of genes according to the time-course transcriptome. Transient (striped square), Plateau (open square), Continuous (filled square), and Late (dotty square) are denoted as all upregulated genes.

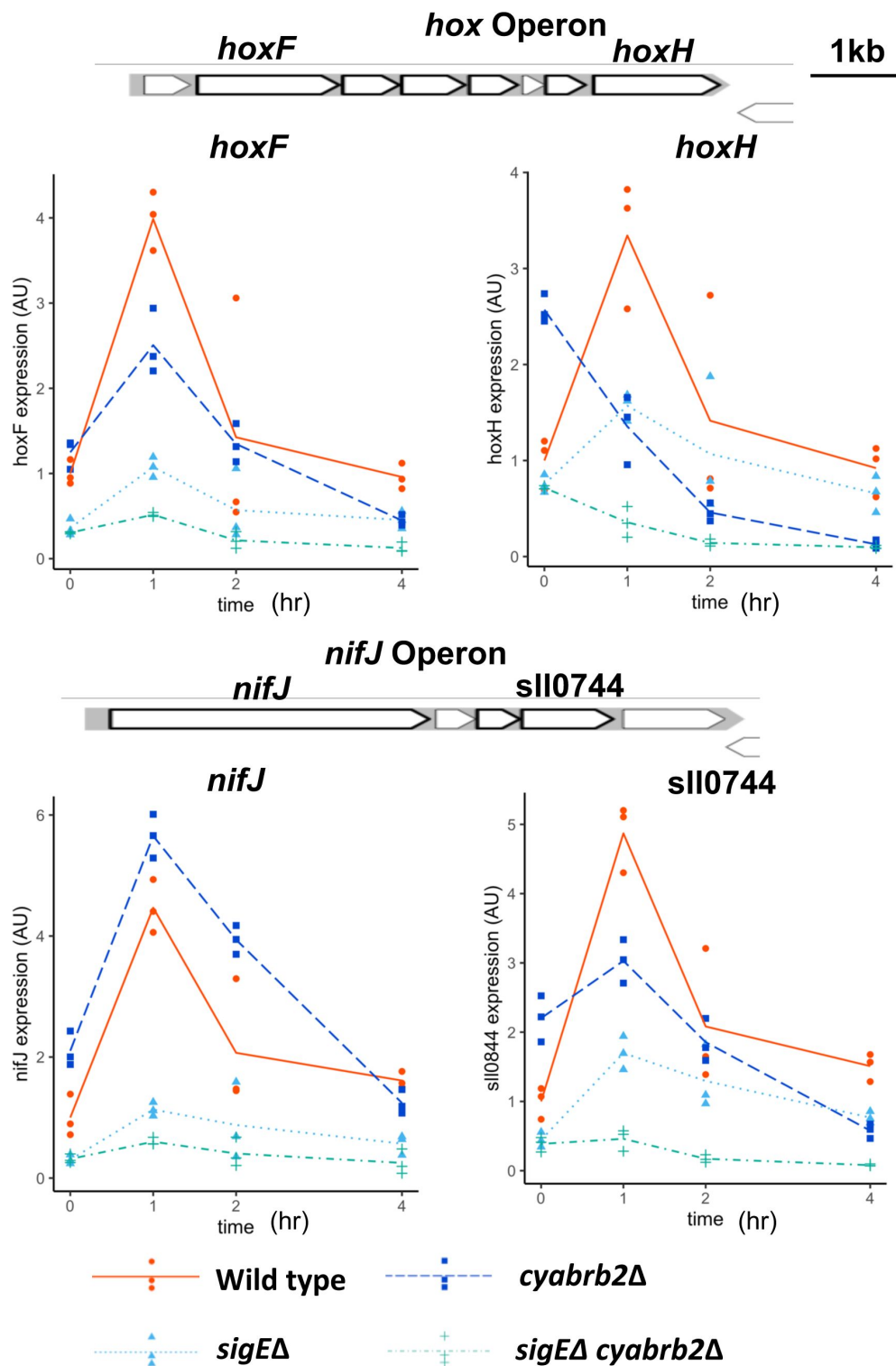


Figure S2

RT-qPCR validated the transiently upregulated genes classified by RNAseq. Transcripts extracted from wildtype (solid line), $\Delta sigE$ mutant (dotted line), $\Delta cyabrb2$ mutant (dashed line), and $\Delta sigE \Delta cyabrb2$ double mutant (dot-dashed line) were assayed in the aerobic condition (0hr) and 1,2,4 hour incubation of microoxic conditions. As the representative of the transiently upregulated genes, expression of *hoxF*, *hoxY*, *nifJ*, and *sll0744* were quantified by RT-qPCR. The line represents the mean of $n=3$, and individual data points are shown as dot plots. Data of each gene is normalized by the mean score of wildtypes in the aerobic condition.

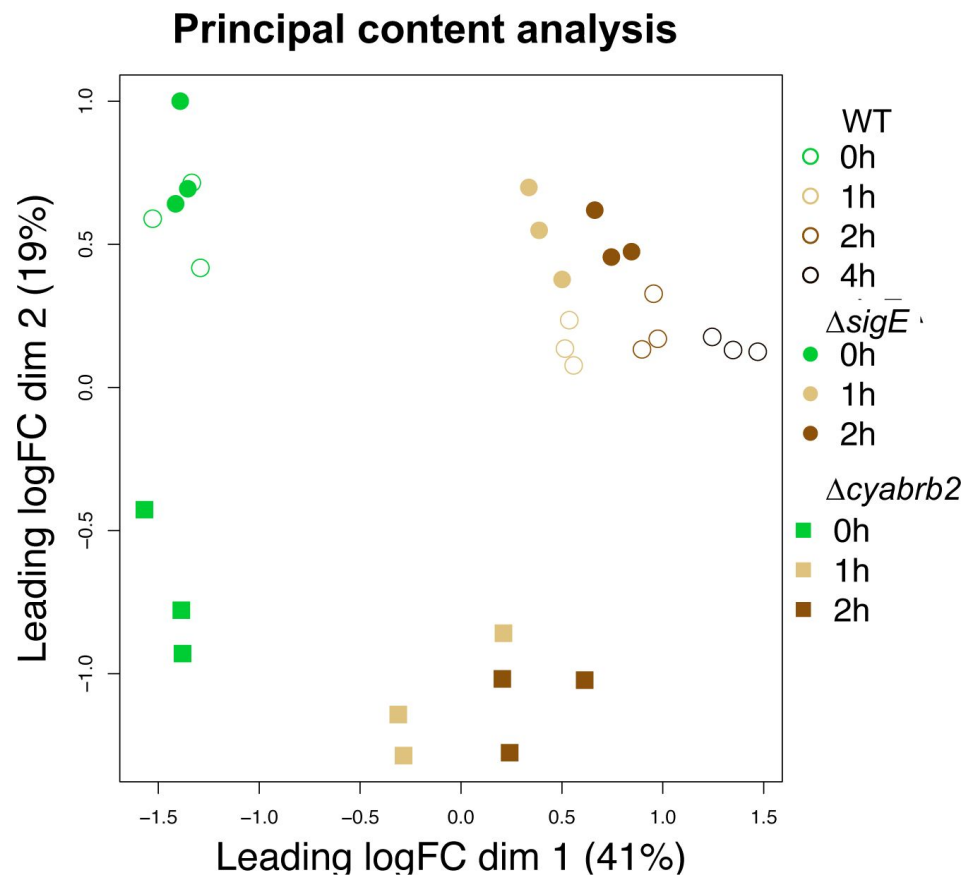


Figure S3

Primary component scatter plot showing the profiles of RNA-seq data.

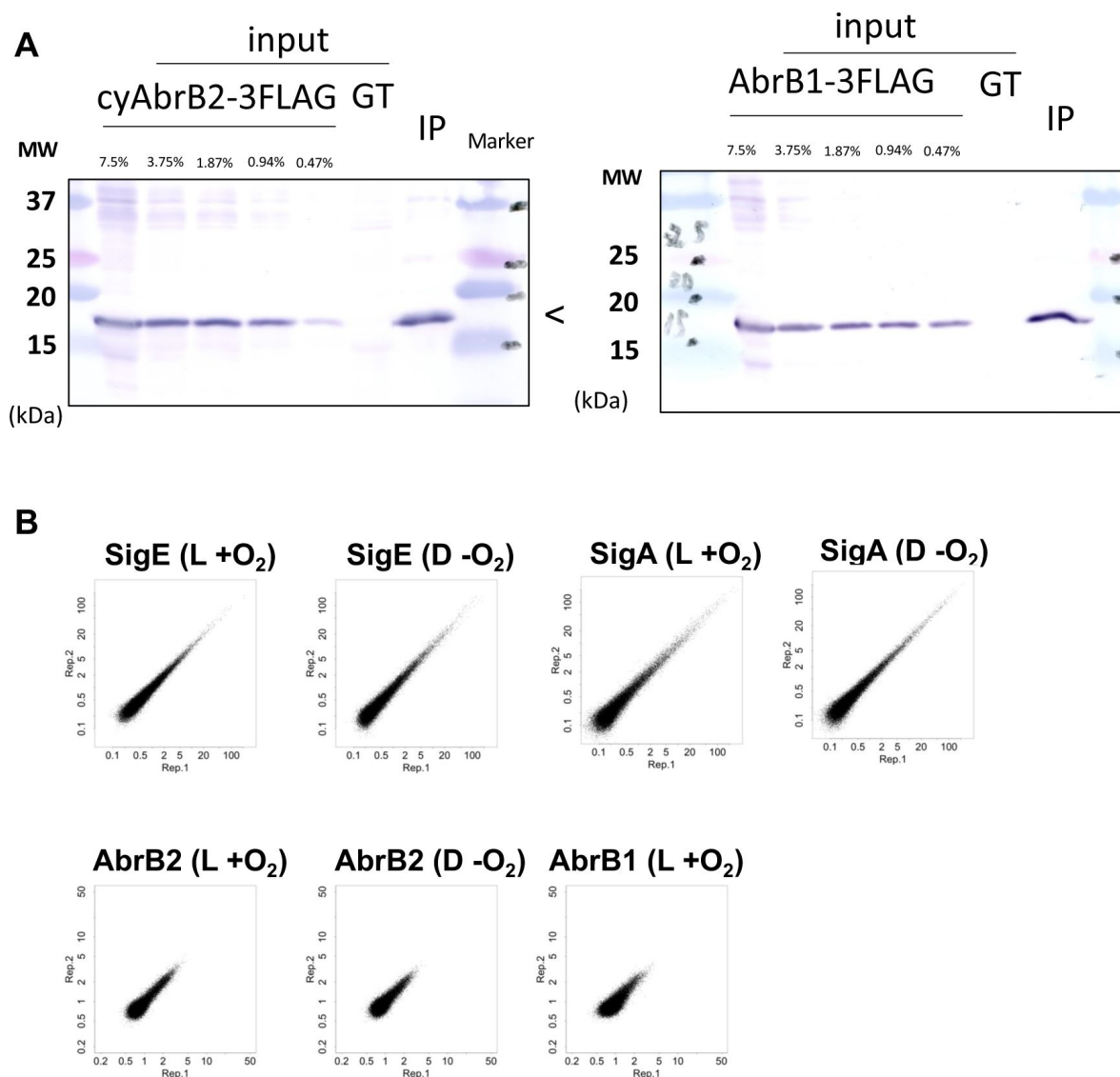


Figure S4

The procedure for ChIP-seq of flag-tagged proteins and Overview of ChIP-seq data

Validation of procedure for ChIP-seq of flag-tagged cyAbrB2, SigE, and SigA. (A) The immunoblot for inputs and immunoprecipitants (IP) of ChIP for cyAbrB2-FLAG and cyAbrB1-FLAG. Input lysate of untagged control (GT) is also loaded. Inputs equivalent to the indicated portion of IP were loaded. (B) Scatter plots showing the reproducibility of two replicates for ChIP-seq assay. ChIP-seq data of SigE, SigA, and cyAbrB2 in aerobic and microoxic conditions and ChIP-seq data of cyAbrB1 in the aerobic condition are shown. Dots indicate normalized IP read count / normalized input read count in each 100bp window. X-axis is the value of replicate1, and Y-axis is the value of replicate 2. (C) and (D) Overview for ChIP-seq of flag tagged cyAbrB2, cyAbrB1, SigE, and SigA. Y-axis indicates [normalized IP read count / normalized input read count at each 25bp window], and X-axis indicates chromosome position. (C) Distribution of cyAbrB2, cyAbrB1, SigE, and SigA across the whole genome of *Synechocystis*. Aerobic (L +O₂) and dark microoxic (D -O₂) data are displayed. (D) Magnified image for chromosome position of 1,550kb-1,800kb. ChIP-seq data of cyAbrB2, cyAbrB1, SigE, and SigA in aerobic and dark microoxic conditions are overlayed. The dots below the graph indicate the binding region of each protein calculated by peak caller.

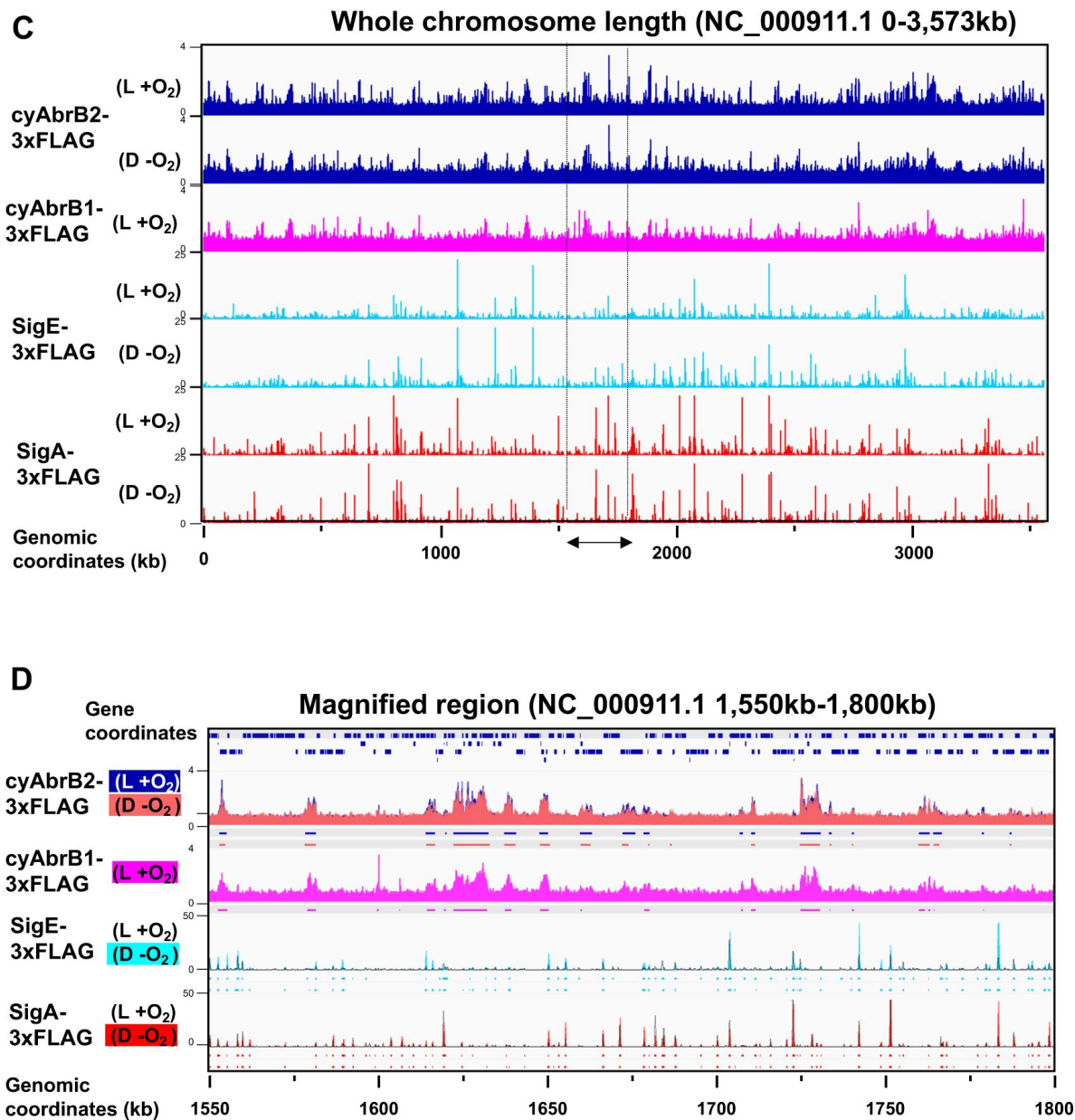


Figure S4 (continued)

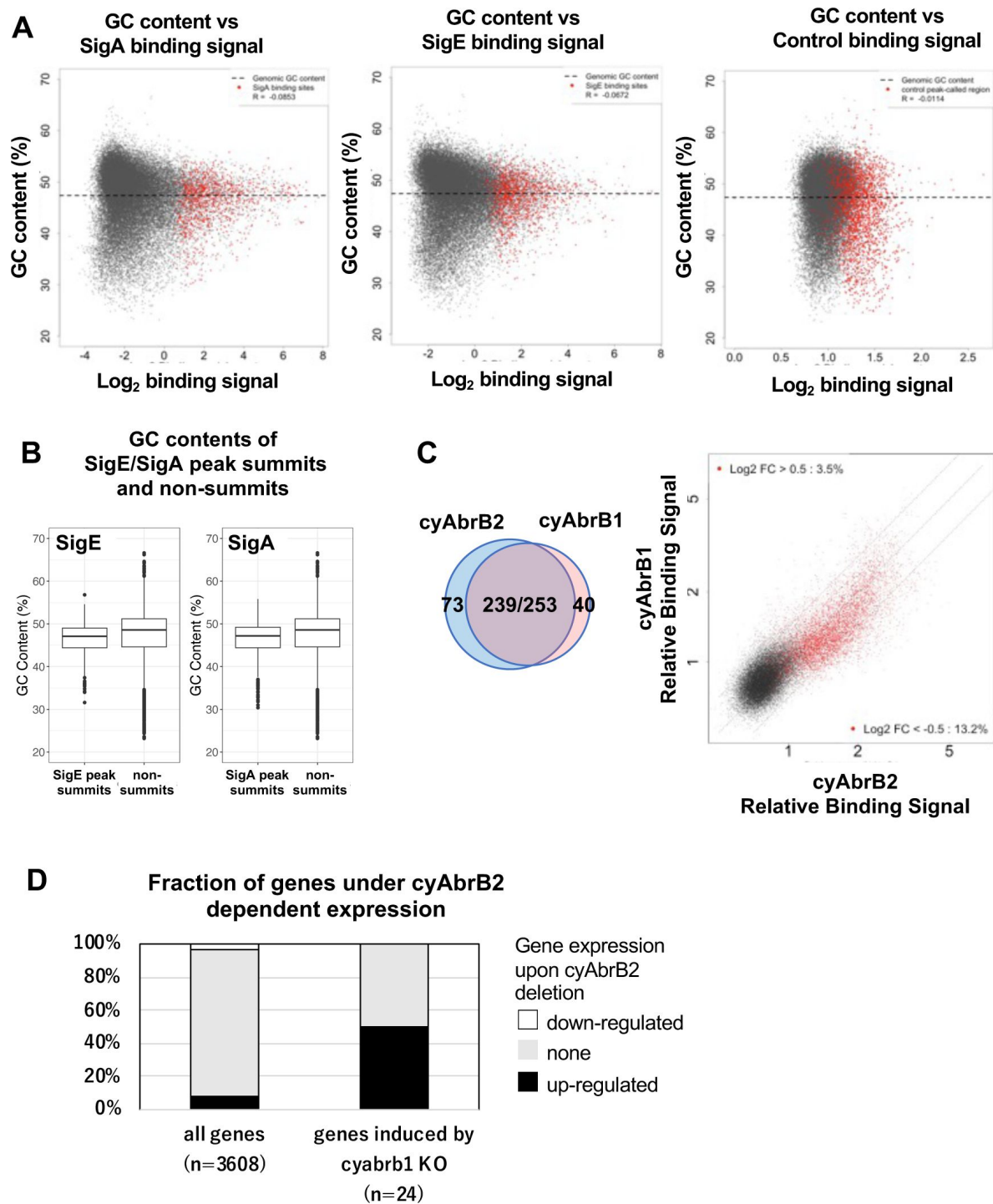


Figure S5

GC content vs ChIP enrichment score of SigA and SigE

(A) Scatter plot showing GC contents in each 100bp vs. binding signal of SigA, SigE, and control IP. Data are displayed as in [Figure 3C](#). (B) GC content in each 100bp of (left) SigE peaks and non-SigE peaks, and (right) SigA peaks and non-SigA peaks. (C) Venn diagram showing overlap of the binding region of cyAbrB1 and cyAbrB2 (left), and scatter plot showing ChIP binding signal of cyAbrB2 (y-axis) and cyAbrB1 (x-axis) in the aerobic condition. Data is plotted as in [Figure 5A](#). (D) Fractions of upregulated and downregulated genes upon the $\Delta cyabr2$ mutant in the aerobic conditions. Fractions of all genes (left $n=3608$) and genes induced by cyAbrB1 knockout (right $n=24$) are shown. Genes induced by cyAbrB1 knockout are from the previous study ([30](#)).

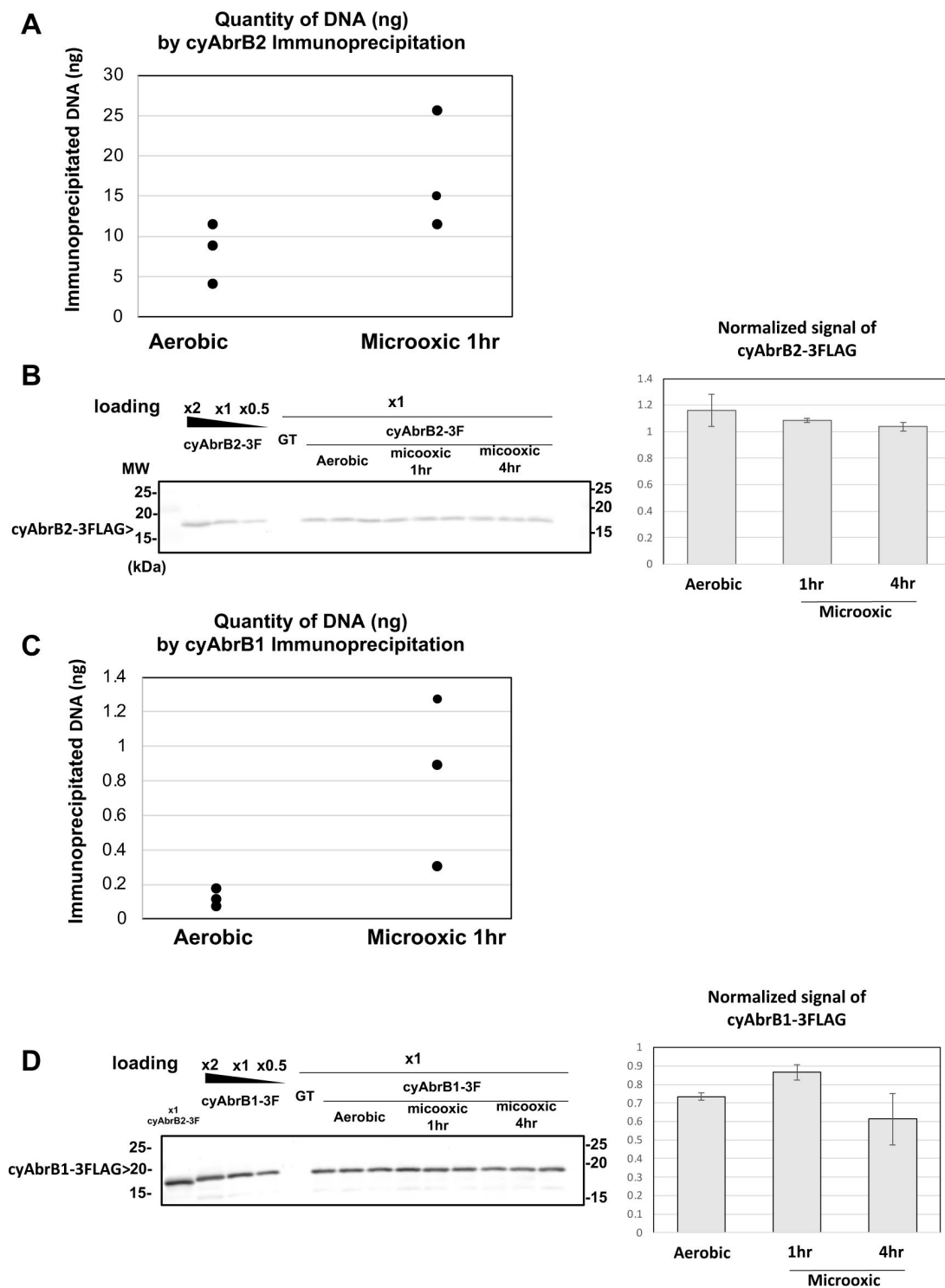


Figure S6

Alteration of cyAbrB2 binding to genome under the microoxic condition

(A) and (C) Amount of precipitated DNA by cyAbrB2 ChIP and cyAbrB1 ChIP, respectively. Three experiments were performed in the aerobic and microoxic conditions. (B) and (D) Western blot images of cyAbrB2-3FLAG and cyAbrB1-3FLAG, respectively. Proteins were extracted in the aerobic condition and 1hr and 4hr incubation under microoxic conditions. The total protein concentration of each sample was adjusted to 4mg/mL, measured by the BCA method. Quantification of cyAbrB2 from western blot image was performed by ImageJ (ver. 2.0.0-rc-65) and plotted in the right graph.

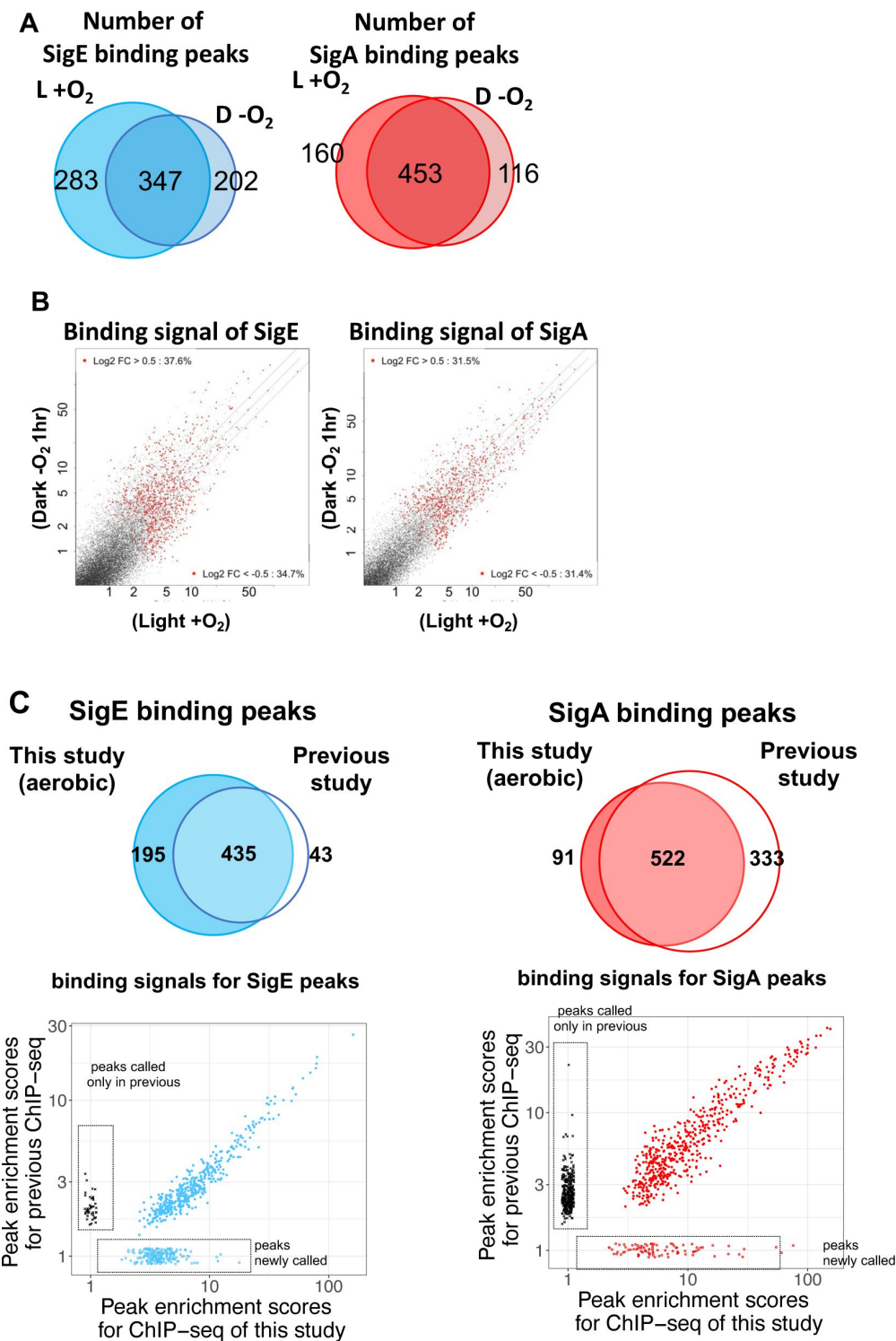


Figure S7

(A) Venn diagram showing the number of peaks of SigE (left) and SigA (right) in aerobic (L + O₂) and dark microoxic (D - O₂) conditions. (B) Scatter plot showing changes in the binding signal of SigE and SigA by 1 hour cultivation under microoxic conditions. The binding signal of each 100 bp window is plotted. (C) Reproducibility of ChIP-seq data of SigA and SigE, compared with the previous study (Kariyazono and Osanai 2022). (top) Venn diagrams show the overlapping of peaks called in this study and the previous study. (bottom) Scatter plot comparing ChIP binding signals of SigA and SigE peaks commonly called in present and previous studies. Plots boxed by dashed lines are peaks called only in the present or previous study.

hox region (Figures 7A, 7B and 7C)

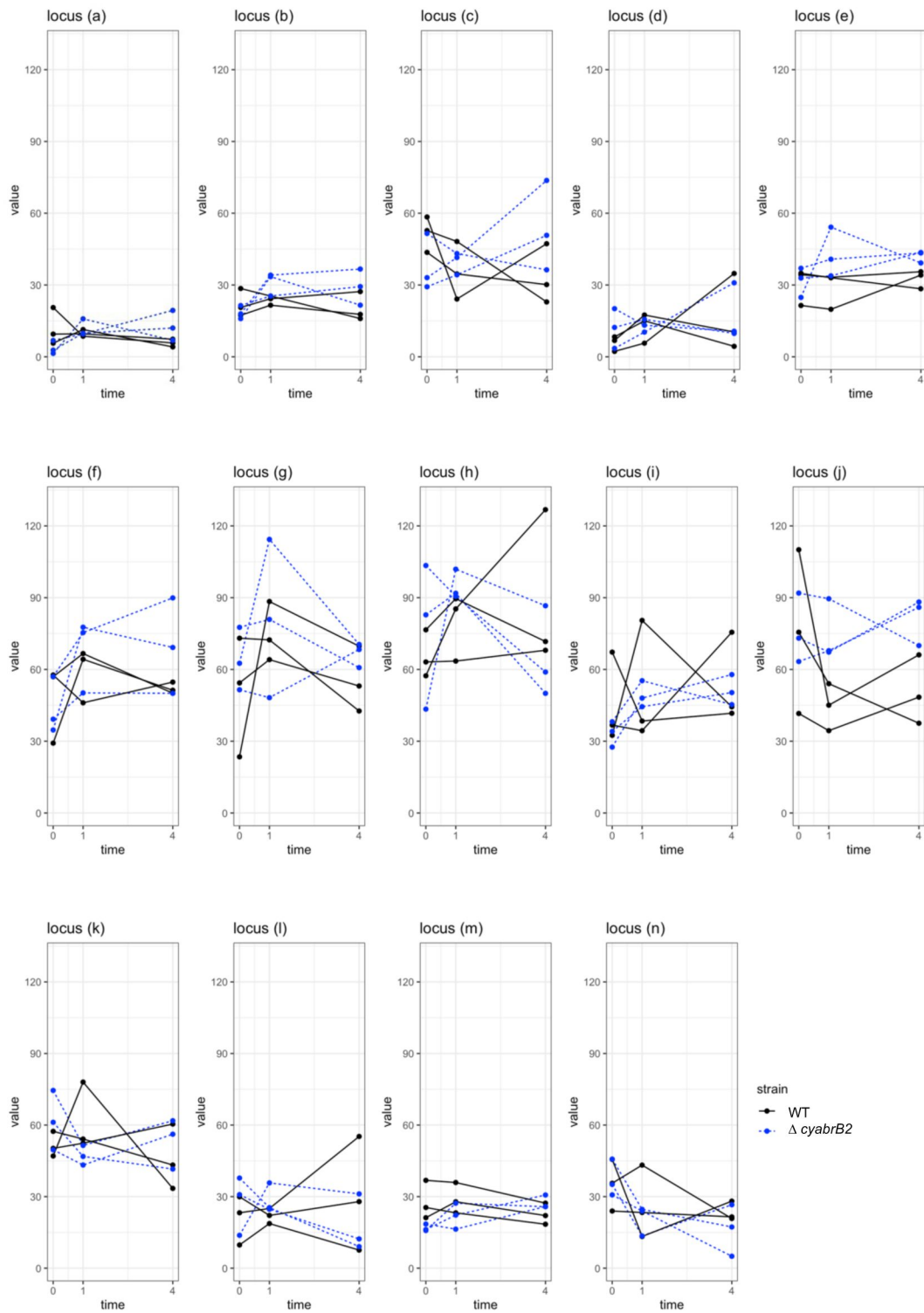


Figure S8

Dynamics of individual 3C scores

Re-plotting of [Figure 5B](#) and [5C](#) with the x-axis showing time (0,1,4 hr in microoxic conditions) and the y-axis showing the interaction frequency. Plots from the individual samples are connected by solid (Wildtype) or dotted ($\Delta cyabrB2$) lines.

nifJ region (Figures 7D, 7E and 7F)

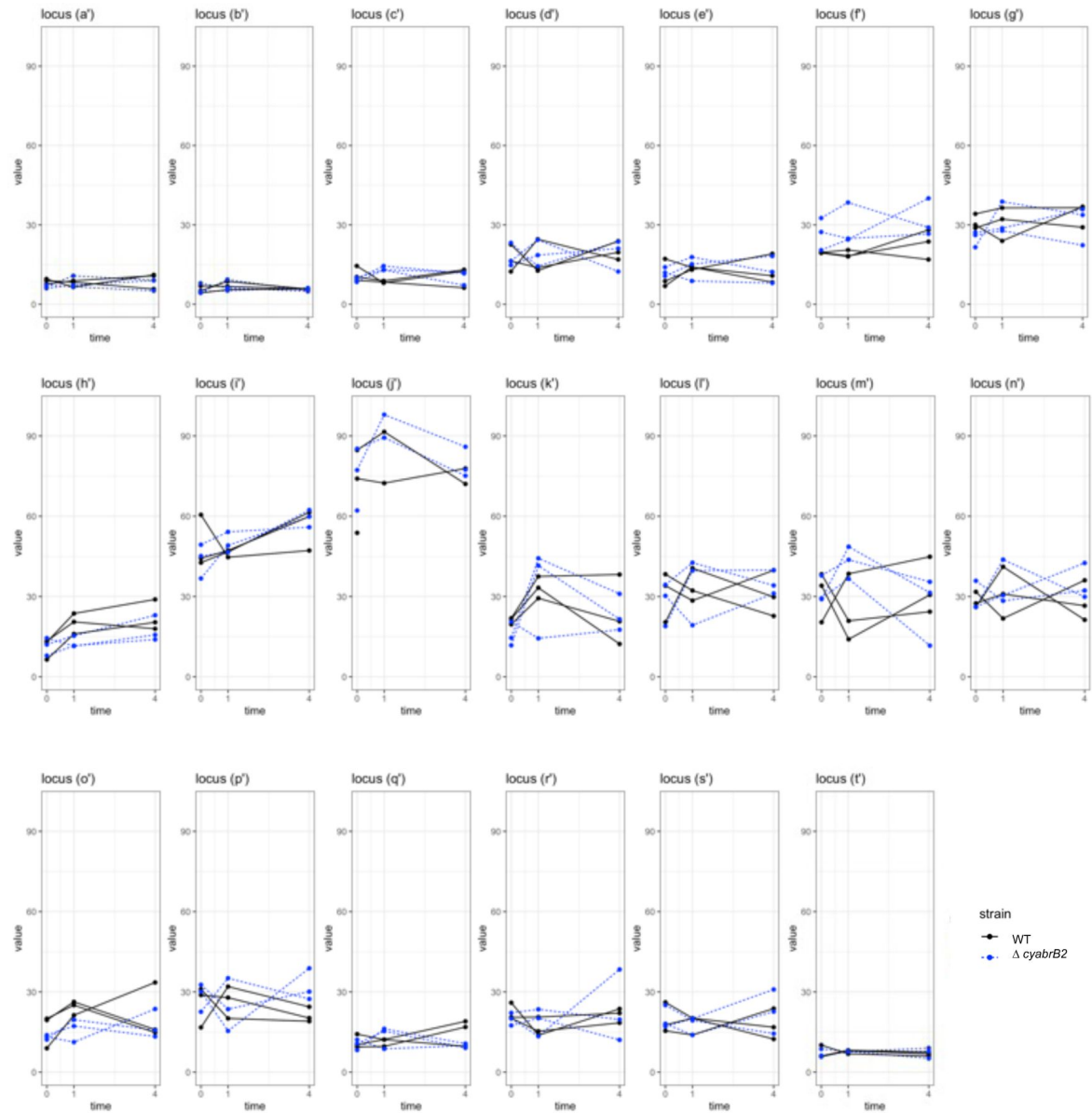


Figure S8 (continued)

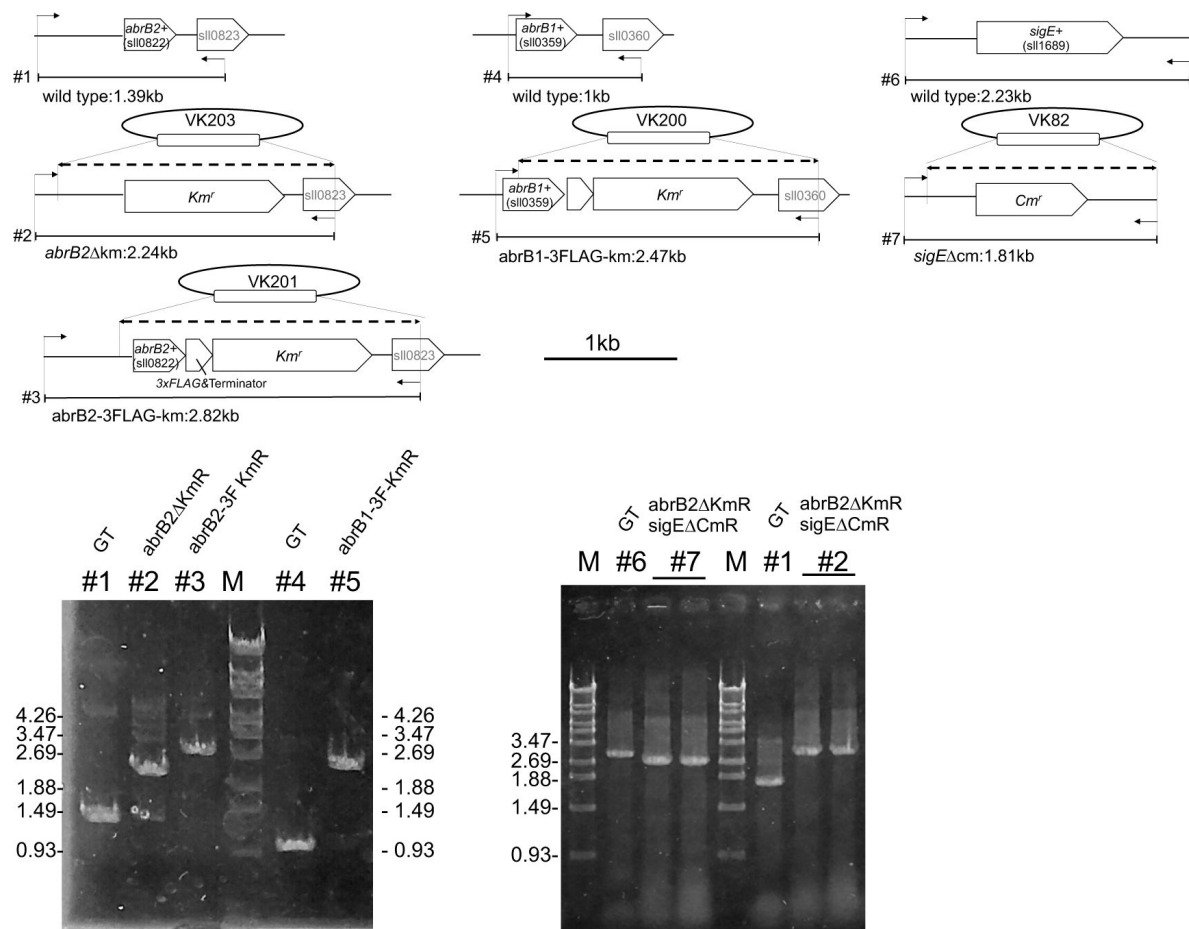
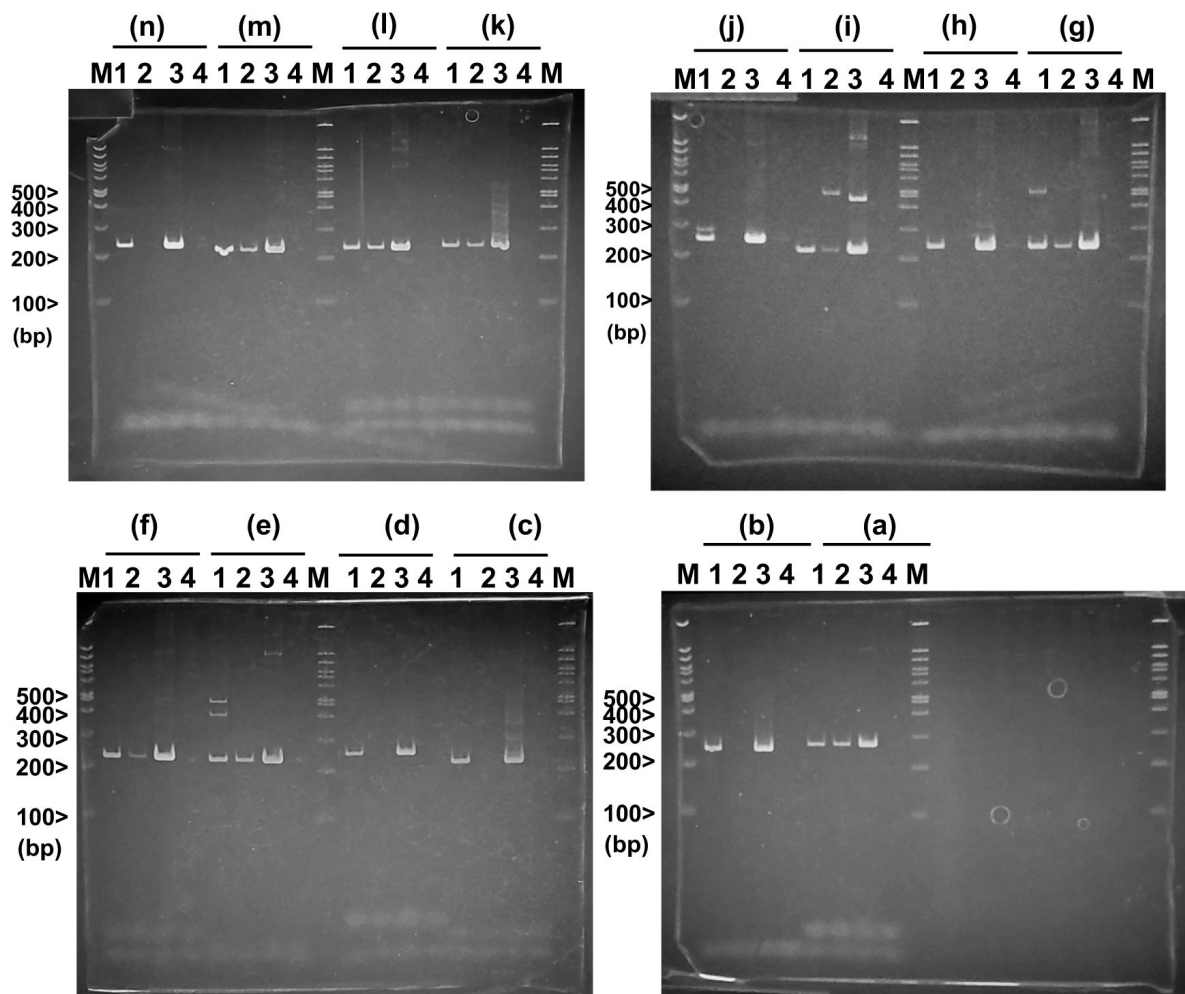


Figure S9

Confirmation of genomic deletion and the epitope tagging of *abrB2* (#1-#3), the epitope tagging of *abrB1* (#4 and #5), and deletion of *sigE* (#6 and #7). Dotted lines are homologous regions between plasmids and the genome. Arrows indicate the position of check primers.

A 3C primers for *hox* operon region



M:100bp ladder (DM2100 SMOBIO)
 1 Calibration sample x500
 2 Calibration sample x2500
 3 3C sample x100
 4 Unligated sample x100

GoTaq® Hot Start
 Program

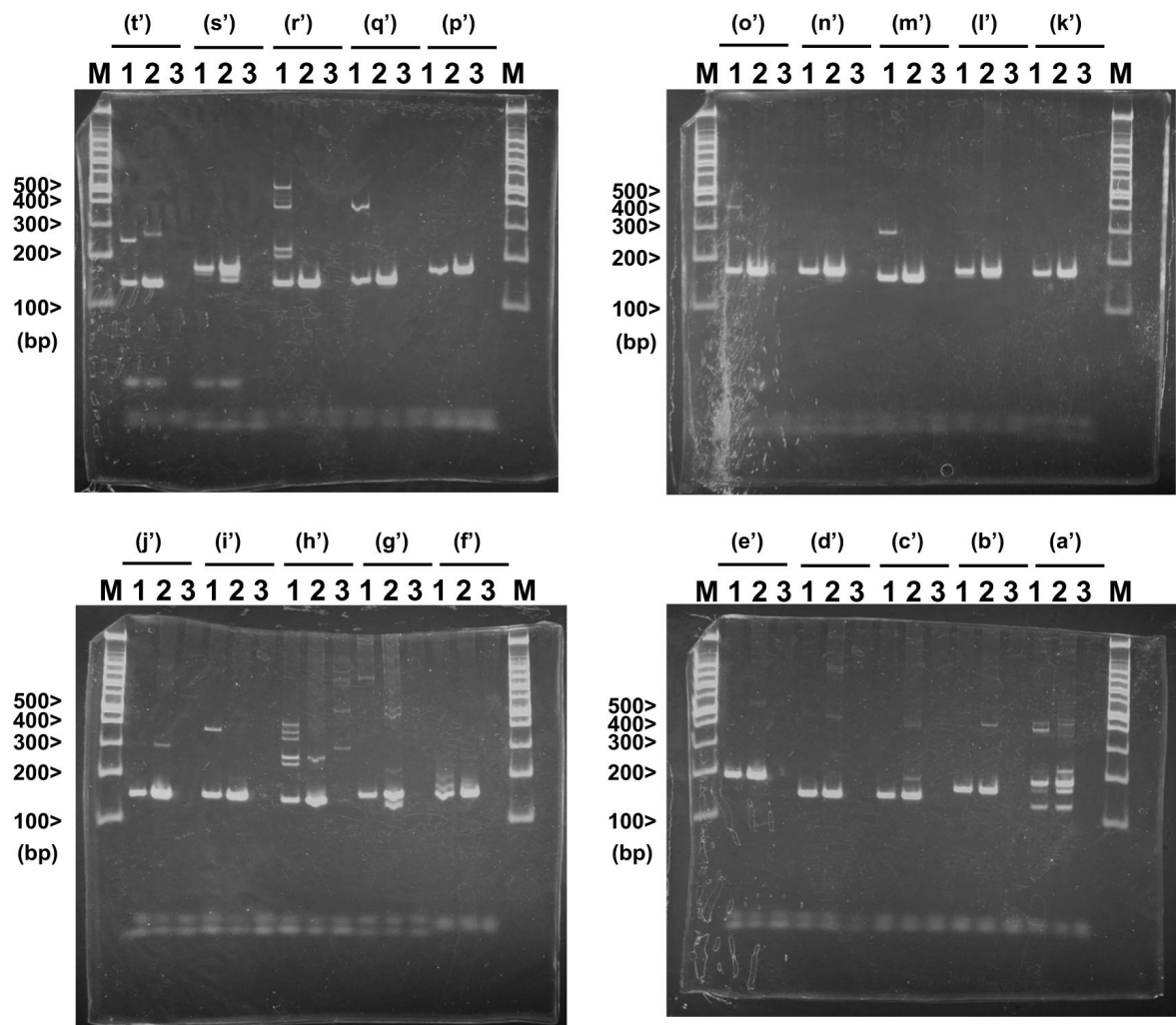
95°C 2min
 95°C 15sec
 60°C 30sec | x35
 4°C Hold

Figure S10

Validation of unidirectional primer sets for 3C assay

The validation of unidirectional primer sets for 3C assay is shown in [Figure 7](#). The 3C sample in this assay is the mixture of all 3C samples assayed in [Figure 7](#).

B 3C primers for *nifJ* region



M:100bp ladder (DM2100 SMOBIO)
1 Calibration sample x1000
2 3C sample x100
3 Unligated sample x100

**GoTaq® Hot Start
Program**

95°C 2min
95°C 15sec
60°C 30sec
4°C Hold

Figure S10 (continued)

Table S1

Fold changes of transcripts from *sigA*, *sigB*, *sigC*, *sigD*, and *sigE*

Sigma factor	locus	0hr vs 1hr		1hr vs 4hr	
		Log2FC	FDR	Log2FC	FDR
SigA	slr0653	-0.873248	0.00766486	-0.0013514	0.99797563
SigB	sll0306	1.38098826	8.42E-06	0.77453605	0.04057775
SigC	sll0184	2.97101055	1.75E-16	1.30743549	0.00067892
SigD	sll2012	0.4701823	0.1498473	-0.4522181	0.32402556
SigE	sll1689	-1.9111759	1.96E-11	-1.1223298	0.00633142

Data is extracted from Supplemental Table S7.

Table S2

List of strains used in this study

Strain ID	Host strain	genotype	Reference
GT			Williams, 1988
G50	GT	$\Delta sigE::KmR$	T. Osanai, et al,2005
KR93	GT	SigA-8His-KmR	(Kariyazono and Osanai 2022)
KR94	GT	SigA-3FLAG-KmR	(Kariyazono and Osanai 2022)
KR340	GT	$\Delta cyabrB2::KmR$	in this study
KR338	GT	cyAbrB(sll0359)-3xFLAG-KmR	in this study
KR339	GT	cyAbrB2(sll0822)-3xFLAG-KmR	in this study
KR359	KR340	$\Delta cyabrB2::KmR \Delta sigE::CmR$	in this study

Table S3

Plasmids used in this study

Plasmid ID		backbone	
VK82	<i>sigE</i> ΔCmR	pTA2 (Toyobo)	transformation for KR359
VK200	AbrB1-3F-KmR	pTA2 (Toyobo)	transformation for KR338
VK201	AbrB2-3F-KmR	pTA2 (Toyobo)	transformation for KR339
VK203	<i>cyabrB2</i> ΔKmR	pTA2 (Toyobo)	transformation for KR340

Oligonucleotides ID		sequence
qPCR_hoxF_F	Figure S2	GTCATCTGCAATGCTGACGAA
qPCR_hoxF_R	Figure S2	CCAACACACTGCGGTCCAT
qPCR_hoxH_F	Figure S2	TCCCGTTACCCGATTGA
qPCR_hoxH_R	Figure S2	GCCCTGGTCGTTGAGGAAA
qPCR_nifJ_sll0741_F	Figure S2	GATTCGGGGCAAAGTTTACG
qPCR_nifJ_sll0741_R	Figure S2	TACCTGGATGGTGAAGTTAG
qPCR_sll0744_F	Figure S2	GCAGCTTAAATGGTTCTACC
qPCR_sll0744_R	Figure S2	CCAAATCTGTGGGTACGTAG
qPCR_nrpB_F	Figure S2	AAAGGGTAAGGGTGCAAAGG
qPCR_nrpB_R	Figure S2	AATTCCTCAAGCGGTTCCAC
new_abrB_0822_del_chkF	Figure S9	AATCCTCCGCCAAACTTTG
abrB_0822_tag4	Figure S9	GACTGATGGTGAAAACCTCC
abrB_0359_chkF	Figure S9	AGAGGGAAAAATTCCTCGGC
abrB_0359_tag4	Figure S9	TCGCCAGGGAAAGTAATTGG
SigE_whole_del_chkF	Figure S9	GGAACGCTTGGAATGGCAAC
sigE_tag4	Figure S9	TGAAGGGGGTAGATTTAAGCC
ChIP_slr2118_F1	Figure 5C #1	CATTTCAAGCGAGATAGCAG
ChIP_slr2118_R1	Figure 5C #2	TGGAAAGTGCCTGTTCAAAG
ChIP_ThoxH_F	Figure 5C #2	CATCACCGAGGGCATATCTAG
ChIP_ThoxH_R	Figure 5C #2	ACATAAGGCTATGGAAACC
ChIP_pHox_2F	Figure 5C #3	ACAGAGGATGGACAAATGTC
ChIP_pHox_2R	Figure 5C #3	TACCGCTTCGTCAATTCTGCT
ChIP_pHox_up_2R	Figure 5C #4	TCGATTACTTCCCTTTAGTG
ChIP_pHox_up_2F	Figure 5C #4	CTTCGGCTCCATGTATTC
ChIP_sll0814_F	Figure 5C #5	GTTACTACCGCTATCTATCG

Table S4

Oligonucleotides used in this study

ChIP_sl0814_R	Figure 5C #5	ATGACGTAAACCGGCTCTGAC
3Chox_Hcommon	Figure 7 (hox region)	CGGTCCCTTTGGACGTAATC
3Chox_H01_pos_+1	Figure 7 (hox region)	GGGTTGGGCGCAACATTAAG
3Chox_H02_pos_+3	Figure 7 (hox region)	CAAAGCGATTTTGGCTGGTG
3Chox_H03_pos_+4	Figure 7 (hox region)	GATGTGGTGGAAGATACGGAC
3Chox_H04_pos_+5	Figure 7 (hox region)	GGAGCCACAGGGGAATTTAAC
3Chox_H05_pos_+6	Figure 7 (hox region)	GGTCAGATTTTATTGGTTGCATG G
3Chox_H06_pos_+7	Figure 7 (hox region)	TTGTCAATGTCAACCGCAAC
3Chox_H07_pos_+8	Figure 7 (hox region)	TTATTGCCCAATCTGGCGAC
3Chox_H08_pos_+11	Figure 7 (hox region)	AGGCAAGACAGATAAACCTTCTG
3Chox_H09_pos_-2	Figure 7 (hox region)	GGGGCATAGGGAGCAAACC
3Chox_H10_pos_-4	Figure 7 (hox region)	GGGGTAGTACCGGAAGTGG
3Chox_H11_pos_-6	Figure 7 (hox region)	TTGGGCTGATTCTTGCAATTC
3Chox_H12_pos_-11	Figure 7 (hox region)	AGCTTTTGTCTAACACCGTATCG
3Chox_H13_pos_-13	Figure 7 (hox region)	AAGCCATCCTTCAAAACCCG
3Chox_H14_pos_-17	Figure 7 (hox region)	GCCCCATCGGTGAGAAAAAG
qPCR_hoxF_F	Figure 7 (hox internal control)	GTCATCTGCAATGCTGACGAA
qPCR_hoxF_R	Figure 7 (hox internal control)	CCAACACACTGCGGTCCAT
3CnifJ_Hcommon	Figure 7 (nifJ region)	ATTCTTTTCGCCAGTTTCTCCAC
3CnifJ_H01_pos-16	Figure 7 (nifJ region)	CATGCTGTTCTGTGCCCATC
3CnifJ_H02_pos-14	Figure 7 (nifJ region)	TCAGTGATATTTTGACCGCTGG
3CnifJ_H03_pos-13	Figure 7 (nifJ region)	AAAACCCCTGCCAACAAGAC
3CnifJ_H04_pos-12	Figure 7 (nifJ region)	CTCCCCACAATAAAGGGCTG
3CnifJ_H05_pos-10	Figure 7 (nifJ region)	TCATAGAAGCGGCGGATACC
3CnifJ_H06_pos-8	Figure 7 (nifJ region)	AGATAACGCCCCGGAACC
3CnifJ_H07_pos-7	Figure 7 (nifJ region)	GTATGGGTCGCCAAAGTACG
3CnifJ_H08_pos-6	Figure 7 (nifJ region)	AGGGATGGGGATCTTTGCTG
3CnifJ_H09_pos-4	Figure 7 (nifJ region)	TTATTGCTTTCACCCCGGC
3CnifJ_H10_pos-3	Figure 7 (nifJ region)	GCCCTAGGCCAAAGAATACC
3CnifJ_H11_pos-1	Figure 7 (nifJ region)	CCAGCCAGGGAGATGACC
3CnifJ_H12_pos+1	Figure 7 (nifJ region)	GCCATTGGGGGTTTATTGGC
3CnifJ_H13_pos+2	Figure 7 (nifJ region)	GTAAACAGATAGGTGGTGCGG

Table S4 (continued)

3CnifJ_H14_pos+6	Figure 7 (nifJ region)	TACAAAGCCAGGTAAGCGTC
3CnifJ_H16_pos+8	Figure 7 (nifJ region)	TCTGACCTATTGCCCTGACC
3CnifJ_H17_pos+9	Figure 7 (nifJ region)	CTGTTGTCATTGCTTGCTCC
3CnifJ_H18_pos+11	Figure 7 (nifJ region)	TGTAGCCAGTGAAGGGGAAG
3CnifJ_H19_pos+14	Figure 7 (nifJ region)	TCCCGGTATTAATGGCATTGG
3CnifJ_H20_pos+15	Figure 7 (nifJ region)	GTTGTGTGATTCCCCAAGCC
3CnifJ_H22_pos+17	Figure 7 (nifJ region)	CTTATTTTGTACCCGGCACCC

Table S4 (continued)

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Reviewer #3 (Public Review):

This work probes the control of the hox operon in the cyanobacterium *Synechocystis*, where this operon directs the synthesis of a bidirectional hydrogenase that functions to produce hydrogen. In assessing the control of the hox system, the authors focused on the relative contributions of cyAbrB2, alongside SigE (and to a lesser extent, SigA and cyAbrB1) under both aerobic and microoxic conditions. In mapping the binding sites of these different proteins, they discovered that cyAbrB2 bound many sites throughout the chromosome, repressed many of its target genes, and preferentially bound regions that were (relatively) rich in AT-residues. These characteristics led the authors to consider that cyAbrB2 may function as a nucleoid-associated protein (NAP) in *Synechocystis*, given the functional similarities with other NAPs like H-NS. They assessed the local chromosome conformation in both wild type and cyabrB2 mutant strains at multiple sites within a 40 kb window on either side of the hox locus, using a region within the hox operon as bait. They concluded that cyAbrB2 functions as a nucleoid associated protein that influences the activity of SigE through its modulation of chromosome architecture.

The authors approached their experiments carefully, and the data were generally very clearly presented. At the same time, the overall work contains many lines of inquiry and different protein investigations that in some ways made it more challenging to identify the overall take-away message(s).

Based on the data presented, the authors make a strong case for cyAbrB2 as a nucleoid-associated protein, given the multiple ways in which it seems to function similarly to the well-studied *Escherichia coli* H-NS protein. They now provide additional commentary that relates cyAbrB2 with other nucleoid-associated proteins.

Previous work had revealed a role for SigE in the control of hox cluster expression, which nicely justified its inclusion (and focus) in this study. The focus on cyAbrB2 is also well-justified, given previous reports of its control of hox expression; however, it shares binding sites with an essential homologue cyAbrB1. Interestingly, while the B1 protein appears to bind similar sites, instead of repressing hox expression, it is known as an activator of this operon. If the information on cyAbrB1 is retained in the manuscript, it would be important to consider how cyAbrB1 activity might influence the results described here (although the authors could also consider removing the cyAbrB1 information to help improve the focus of the manuscript).

<https://doi.org/10.7554/eLife.94245.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife assessment

The authors provide solid data on a functional investigation of potential nucleoid-associated proteins and the modulation of chromosomal conformation in a model cyanobacterium. While the experiments presented are convincing, the manuscript could benefit from restructuring towards the precise findings; alternatively, additional data buttressing the claims made would significantly enhance the study. These valuable findings will be of interest to the chromosome and microbiology fields.

We appreciate editors for taking time for assessment and reviewers for giving critical suggestions. Both reviewers were concerned about our interpretation of 3C data, and Reviewer #2 suggested the biochemistry of cyAbrB2 to reinforce our claim. We agree with the concern and suggest editors add a sentence “How cyAbrB2 affects chromosome structure is still elusive from this study, and the biochemical assays are needed in the future experiment.” to the eLife assessment.

The major revision points are the following;

Reconstruction of Figures

Previous Figure 5E has been omitted

Additional 3C data on the nifJ region

Rephrasing the conclusion of 3C data

Additional discussion on cyAbrB2 and NAPs

Reviewer #1 (Public Review):

Strength:

At first glance, I had a very positive impression of the overall manuscript. The experiments were well done, the data presentation looks very structured, and the text reads well in principle.

Weakness:

Having a closer look, the red line of the manuscript is somewhat blurry. Reading the abstract, the introduction, and parts of the discussion, it is not really clear what the authors exactly aim to target. Is it the regulation of fermentation in cyanobacteria because it is under-investigated? Is it to bring light to the transcriptional regulation of hydrogenase genes? The regulation by SigE? Or is it to get insight into the real function of cyAbrB2 in cyanobacteria? All of this would be good of course. But it appears that the authors try to integrate all these aspects, which in the end is a little bit counterintuitive and in some places even confusing. From my point of view, the major story is a functional investigation of the presumable transcriptional regulator cyAbrB2, which turned out to be a potential NAP. To demonstrate/prove this, the hox genes have been chosen as an example due to the fact that a regulatory role of cyAbrB2 has already been described. In my eyes, it would be good to restructure or streamline the introduction according to this major outcome.

As you pointed out, the major focus of this study is cyAbrB2 as a potential NAPs. To focus on NAPs, we simplified the first paragraph of the discussion (ll.246-263) and added the section comparing cyAbrB2 with other known NAPs (ll.269-299). To emphasize the description of cyAbrB2, we also rearranged the figures and divided the analysis on cyAbrB2 ChIP into two figures. We reduced the first paragraph of the introduction but mostly preserved the

composition of the introduction to keep the general to specific pattern, even though the manuscript is blurry.

Points to consider:

The authors suggest that the microoxic condition is the reason for the downregulation of e.g. photosynthesis (l.112-114). But of course, they also switched off the light to achieve a microoxic environment, which presumably is the trigger signal for photosynthesis-related genes. I suggest avoiding making causal conclusions exclusively related to oxygen and recommend rephrasing (for example, "were downregulated under the conditions applied").

We agree with this point. We rephrased l.114 to "by the transition to dark microoxic conditions from light aerobic conditions" (ll.108-109).

The authors hypothesized that cyAbrB2 modulates chromosomal conformation and conducted a 3C analysis. But if I read the data in Figure 5B & C correctly, there is a lot of interaction in a range of 1650 and 1700 kb, not only at marked positions c and j. Positions c and j have been picked because it appears that cyAbrB2 deletion impacts this particular interaction. But is it really significant? In the case of position j the variation between the replicates seems quite high, in the case of position c the mean difference is not that high. Moreover, does all this correlate with cyAbrB2 binding, i.e. with positions of gray bars in panel A? If this was the case, the data obtained for the cyabrB2 mutant should look totally different but they are quite similar to WT. That's why the sentence "By contrast, the interaction frequency in ΔcyabrB2 mutant were low and unchanged in the aerobic and microoxic conditions" does not fit to the data shown. But I have to mention that I am not an expert in these kinds of assays. Nevertheless, if there is a biological function that shall be revealed by an experiment, the data must be crystal clear on that. At least the descriptions of the 3C data and the corresponding conclusions need to be improved. For me, it is hard to follow the authors' thoughts in this context.

According to your suggestion, we again have carefully observed the 3C data. Furthermore, we conducted an additional 3C experiment on nifJ region (Figures 7F-J). Then we admit we had overinterpreted the 3C data. Therefore, we rewrote the result and discussion of the 3C assay in line with the data (ll.220-245) and removed the previous Figure 5E. Following are individual responses.

Positions c and j have been picked because it appears that cyAbrB2 deletion impacts this particular interaction. But is it really significant?

We could not find statistically significant differences at locus c and j. Therefore, we added this in the result section "Note that the interaction scores exhibit considerable variability and we could not detect statistical significance at those loci." (ll.231-232)

does all this correlate with cyAbrB2 binding, i.e. with positions of gray bars in panel A?

As you are concerned, interaction frequency and cyAbrB2 binding do not correlate. Therefore, we withdraw the previous claim and stated as follows; "Moreover, our 3C data did not support bridging at least in hox region and nifJ region, as the high interaction locus and cyAbrB2 binding region did not seem to correlate (Figure 7)." (ll.280-282)

If this was the case, the data obtained for the cyabrB2 mutant should look totally different but they are quite similar to WT.

We rewrote it as follows; “Then we compared the chromatin conformation of wildtype and cyabrb2Δ. Although overall shapes of graphs did not differ, some differences were observed in wildtype and cyabrb2Δ (Figures 7B and 7G); interaction of locus (c) with hox region were slightly lower in cyabrb2Δ and interaction of loci (f') and (g') with nifJ region were different in wildtype and cyabrb2Δ. Note that the interaction scores exhibit considerable variability and we could not detect statistical significance at those loci.” (l1.228-232)

That's why the sentence "By contrast, the interaction frequency in Δcyabrb2 mutant were low and unchanged in the aerobic and microoxic conditions" does not fit to the data shown.

We rewrote the sentence as follow; “While the interaction scores exhibit considerable variability, the individual data over time demonstrate declining trends of the wildtype at locus (c) and (j) (Figure S8). In Δcyabrb2, by contrast, the interaction frequency of loci (c) and (j) was unchanged in the aerobic and microoxic conditions (Figure 7E). The interaction frequency of locus (c) in Δcyabrb2 was as low as that in the microoxic condition of wildtype, while that of locus (j) in Δcyabrb2 was as high as that in the aerobic condition of wildtype (Figures 7B and 7C).” (l1.238-243)

The figures are nicely prepared, albeit quite complex and in some cases not really supportive of the understanding of the results description. Moreover, they show a rather loose organization that sometimes does not fit the red line of the results section. For example, Figure 1D is not mentioned in the paragraph that refers to several other panels of the same figure (see lines 110-128). Panel 1D is mentioned later in the discussion. Does 1D really fit into Figure 1 then? Are all the panels indeed required to be shown in the main document? As some elements are only briefly mentioned, the authors might also consider moving some into the supplement (e.g. left part of Figure 1C, Figure 2A, Figure 3B ...) or at least try to distribute some panels into more figures. This would reduce complexity and increase comprehensibility for future readers. Also, Figure 3 is a way too complex. Panel G could be an alone-standing figure. The latter would also allow for an increase in font sizes or to show ChIP data of both conditions (L+O2 and D-O2) separately. Moreover, a figure legend typically introduces the content as a whole by one phrase but here only the different panels are described, which fits to the impression that all the different panels are not well connected. Of course, it is the decision of the authors what to present and how but may they consider restructuring and simplifying.

According to the advice, we have rearranged the Figure composition.

The left side of Figure 1C has been moved to supplement. Instead, representative expression fold changes of “Transient”, “Plateau”, “Continuous”, and “Late” genes are shown for comprehensibility. We left Figure 1D in Figure 1, as this diagram shows our motive to focus on hox and nifJ. We moved Figure 2A to supplement. We did not move Fig3B, as this figure shows the distribution of cyAbrB2 (“long tract of AT-rich DNA”) comprehensively and simply. We agree that Figure 3 was too complex. Therefore, we moved Figures 3F and 3G to a new independent figure (Figure 4). In Figure 4C (former 3G), we show the ChIP data of the L+O2 condition only, and the change of ChIP data under the D-O2 condition is shown in Figure 5. The schematic image showing cyanobacterial chromosome and NAPs (previous Figure 5E) was omitted because it was overinterpreting.

The authors assume a physiological significance of transient upregulation of e.g. hox genes under microoxic conditions. But does the hydrogenase indeed produce hydrogen under the conditions investigated and is this even required? Moreover, the authors use the term "fermentative gene". But is hydrogen indeed a fermentation product, i.e. are

protons the terminal electron acceptor to achieve catabolic electron balance? Then huge amounts of hydrogen should be released. Comment should be made on this.

This is a very important point; Yes, hydrogenase indeed produces hydrogen under the conditions we investigated, and proton accepts a majority of reducing power under the dark microoxic condition. We wrote in the introduction section as follows; “Hydrogen is generated in quantities comparable to lactate and dicarboxylic acids as the result of electron acceptance in the dark microoxic condition (Akiyama and Osanai 2023; Iijima et al. 2016)” (ll.54-55). The detailed explanation is below, although omitted from the manuscript.

A recent study (Akiyama and Osanai 2023) quantified the consumed glycogen and secreted fermentative products (hydrogen, lactate, dicarboxylic acid, and acetate) in the *Synechocystis* under the dark microoxic condition, the same conditions as we investigated. The system of the study consists of a 10 mL liquid layer and a 10 mL gas layer, cultivated for 3 days under dark microoxic conditions. Then the amounts of lactic acid, dicarboxylic acid, and hydrogen were approximately 2 μmol , 3.5 μmol , and 11 μmol (assuming the gas layer was at 1 atm and ignoring aqueous population), respectively. On the other hand, glycogen equivalent to 15 μmol of glucose was consumed in the system. This estimate supports hydrogen accounts for a substantial portion of fermentative products during dark microoxic conditions.

The necessity of hydrogen production under dark microoxic conditions was demonstrated in (Gutekunst et al. 2014). They show hydrogenase activity is required for the mixotrophic growth in the light-dark and microoxic cycle with arginine. The necessity remains unclear in our conditions because we only performed continuous dark microoxic conditions without glucose.

The authors also mention a reverse TCA cycle. But is its existence an assumption or indeed active in cyanobacteria, i.e. is it experimentally proven? The authors are a little bit vague in this regard (see lines 241-246).

We misused the Terminology. We mean to mention the “reductive branch of TCA”. Cyanobacteria conduct the branched TCA cycle under microoxic conditions. One of the branches is the reductive branch, which reduces oxaloacetate to produce malate. We corrected “reverse TCA cycle” to “reductive branch of TCA”. (Figure 1D and ll.260-262)

Reviewer #2 (Public Review):

*This work probes the control of the hox operon in the cyanobacterium *Synechocystis*, where this operon directs the synthesis of a bidirectional hydrogenase that functions to produce hydrogen. In assessing the control of the hox system, the authors focused on the relative contributions of cyAbrB2, alongside SigE (and to a lesser extent, SigA and cyAbrB1) under both aerobic and microoxic conditions. In mapping the binding sites of these different proteins, they discovered that cyAbrB2 bound many sites throughout the chromosome repressed many of its target genes, and preferentially bound regions that were (relatively) rich in AT-residues. These characteristics led the authors to consider that cyAbrB2 may function as a nucleoid-associated protein (NAP) in *Synechocystis*, given its functional similarities with other NAPs like H-NS. They assessed the local chromosome conformation in both wild-type and cyabrB2 mutant strains at multiple sites within a 40 kb window on either side of the hox locus, using a region within the hox operon as bait. They concluded that cyAbrB2 functions as a nucleoid-associated protein that influences the activity of SigE through its modulation of chromosome architecture.*

The authors approached their experiments carefully, and the data were generally very clearly presented and described.

Based on the data presented, the authors make a strong case for cyAbrB2 as a nucleoid-associated protein, given the multiple ways in which it seems to function similarly to the well-studied Escherichia coli H-NS protein. It would be helpful to provide some additional commentary within the discussion around the similarities and differences of cyAbrB2 to other nucleoid-associated proteins, and possible mechanisms of cyAbrB2 control (post-translational modification; protein-protein interactions; etc.). The manuscript would also be strengthened with the inclusion of biochemical experiments probing the binding of cyAbrB2, particularly focusing on its oligomerization and DNA polymerization/bridging potential.

We agree with the comment that the biochemical experiments will deepen our insights into the cyAbrB2 and chromatin conformation. As the reviewer pointed out, the biochemical assay will provide valuable information on mechanisms of cyAbrB2 control, such as post-transcriptional modification, cooperation with cyAbrB1, oligomerization, and the structure of cyAbrB2-bound DNA. However, we think those potential findings are worth of new independent research paper, rather than a part of this paper. Therefore, we added a discussion mentioning biochemistry as the future work (ll.275-290; the section of “The biochemistry of cyAbrB2 will shed light on the regulation of chromatin conformation in the future”).

Previous work had revealed a role for SigE in the control of hox cluster expression, which nicely justified its inclusion (and focus) in this study. However, the results of the SigA studies here suggested that SigA both strongly associated with the hox promoter, and its binding sites were shared more frequently than SigE with cyAbrB2. The focus on cyAbrB2 is also well-justified, given previous reports of its control of hox expression; however, it shares binding sites with an essential homologue cyAbrB1. Interestingly, while the B1 protein appears to bind similar sites, instead of repressing hox expression, it is known as an activator of this operon. It seems important to consider how cyAbrB1 activity might influence the results described here.

We infer that the minor side of the bimodal SigE peak is the genuine population that contributes to hox transcription, as hox genes are expressed in a SigE-dependent manner (Figure S2). We considered the strong SigA peak upstream of the hox operon binds the promoter of TU1715, the opposite direction of the hox operon. We added a description of the single SigA peak and bimodal SigE peak near the TSS of the hox operon as follows;

“A bimodal peak of SigE was observed at the TSS of the hox operon in a microoxic-specific manner (Figure 6C bottom panel). The downstream side of the bimodal SigE peak coincides with SigA peak and the TSS of TU1715. Another side of the bimodal peak lacked SigA binding and was located at the TSS of the hox operon (marked with an arrow in Figure 6C), although the peak caller failed to recognize it as a peak.” (ll.206-209)

The point that cyAbrB1 binds similar sites as cyAbrB2, despite regulating hox expression in the opposite direction, is very interesting. Therefore, we referred to the transcriptome data of the cyAbrB1 knockdown strain and compared the impact of cyAbrB1 knockdown and cyAbrB2 deletion. We described in result and discussion as follows;

“we referred to the recent study performing transcriptome of cyAbrB1 knockdown strain, whose cyAbrB1 protein amount drops by half (Hishida et al. 2024). Among 24 genes induced by cyAbrB1 knockdown, 12 genes are differentially downregulated genes in cyabrB2Δ in our study (Figure S5D).” (ll.162-165)

“CyAbrB1, the homolog of cyAbrB2, may cooperatively work, as cyAbrB1 directly interacts with cyAbrB2 (Yamauchi et al. 2011), their distribution is similar, and they partially share their target genes for suppression (Figures 3A S5C and S5D). The possibility of cooperation

would be examined by the electrophoretic mobility shift assay of cyAbrB1 and cyAbrB2 as a complex. Despite their similar repressive function, cyAbrB1 and cyAbrB2 regulate *hox* expression in the opposite directions, and their mechanism remains elusive.” (l1.292-296)

Hox operon differs from this general tendency. To see if cyAbrB1 behaves differently from cyAbrB2 in the *hox* operon, we did an additional ChIP-qPCR experiment on cyAbrB1 in the aerobic condition and the dark microoxic condition (Figure 5C). However, we could not find the difference.

Reviewer #1 (Recommendations For The Authors):

Figure 1B: I recommend changing the header in the grey bar to terms like "upregulated" and "downregulated", which are also used in the legend description. Upregulation of genes can also be a result of de-repression, which is why the term "activated" is somewhat misleading.

Corrected.

Lines 114-116: It is unclear what the authors exactly mean here. Please clarify.

We rephrase the sentence “The enrichment in the butanoate metabolism pathway indicates the upregulation of genes involved in carbohydrate metabolism. We further classified genes according to their expression dynamics.” (l1.110-111)

Reviewer #3 (Recommendations For The Authors):

Major/experimental comments:

(1) For the chromosome conformation capture experiments, it is indicated that these were conducted at aerobic (1hr) and microoxic (4 hr) conditions. But the data presented in Figure 1 suggest that 1 hr corresponds to the beginning of microoxic growth, and that time 0 is aerobic. The composite 3C data in Figure 5 show some interesting but specific differences. It is appreciated that the authors presented the profiles for individual samples in Figure S7, and the differences here do not seem to be as compelling. Are the major differences being highlighted significantly (statistically) different (e.g. at the (c) and (j) loci)? Might the differences be starker if an earlier aerobic condition (e.g. time 0) had been used instead of the 1 hr - microoxic - timepoint?

Previous Figure 5 consisted of three time points (solid line: aerobic condition, dashed line: 1hr of microoxic condition, and dotted line: 4hr of microoxic condition). We omitted data of 4hr in the main figure (Figure 7) as 4hr in microoxic conditions makes data complicated. Three time points are shown in the profiles of individual loci (Figure S8).

There is no statistical significance found in (c) and (j) loci by t-test. Therefore, we have toned down the interpretation of 3C data as follows; “Our 3C result demonstrated that cyAbrB2 influences the chromosomal conformation of *hox* and *nifj* region to some extent (Figure 7).” (l1.325-326)

*(2) This is a complicated system that involves multiple regulatory proteins, each of which is differentially affected by the growth conditions (aerobic/microoxic). It is obviously beyond the scope of this work to probe deeply into all of these proteins. The focus here was on cyAbrB2, and to a slightly lesser extent SigE; however, based on the data presented, it seems that SigA and cyAbrB1 may be equally important contributors to *hox* control/expression, and in the case of cyAbrB1, possibly also to chromosome conformation. cyAbrB1 appears to have the same binding sites as cyAbrB2, and has been reported to interact with cyAbrB2. Given this association, it is possible that the two*

proteins may affect the binding of each other, and that loss of one might lead to enhanced binding by the other (or binding may require heterooligomerization?). Probing the regulatory interplay between these two proteins (or at least discussing it) feels important. Conducting e.g. mobility shift assays with each protein, both individually and together, could possibly allow for some understanding of how they function together.

We agree that the biochemistry of cyAbrB2 and cyAbrB1 may explain why cyAbrB1 and cyAbrB2 bind long tracts of AT-rich genome regions in vitro. We would like to put the biochemistry future plan as we think biochemistry data is beyond the present study.

The idea that cyAbrB1 and cyAbrB2 cooperate to form heterooligomers and broad binding to the genome is a very rational and interesting prediction. We add this idea to the discussion “Overall, the biochemistry integrating assay conditions (PTM, buffer condition, and cooperation with cyAbrB1) and output (DNA binding, oligomerization, and DNA structure) will deepen the understanding of cyAbrB2 as cyanobacterial NAPs.”(l.287-290). We also compared our transcriptome of Δ _cyabrb2 with the recent study of cyabrb1 knockdown (l. 162-165), and concluded “they partially share their target genes for suppression (Figures 3A S5C and S5D)” (l. 293).

(3) Throughout the manuscript, there is reference made to cyAbrB2 binding becoming 'blurry' or non-specific under microoxic conditions. It is not clear what this means. It appears that when cyAbrB2 binds, any given protected region can be quite extensive, which can be suggestive of polymerization along the chromosome. Are the boundaries for binding sites typically clearly delineated, and this changes when the cultures are growing under microoxic conditions? There is also no mention made anywhere about oligomerization potential for cyAbrB2, which would be important for the polymerization, and bridging suggested for cyAbrB2 in the model presented in Figure 5. Previous publications (Song et al., 2022; Ishi et al., 2008) have suggested that it can exist as a dimer in vivo, but that in vitro it is largely monomeric. The manuscript would benefit from some additional biochemical analyses of cyAbrB2 binding activity, with a particular focus on DNA binding and oligomerization/bridging potential, and some additional discussion about these characteristics as well.

Throughout the manuscript, there is reference made to cyAbrB2 binding becoming 'blurry' or non-specific under microoxic conditions. It is not clear what this means.

In order to clearly describe “cyAbrB2 binding becomes blurry”, we rearranged the figure composition and made an exclusive figure (Figure 5). We also rephrased the description by adopting the reviewer’s word “boundaries for binding sites”, as this phrase well describes the change. “When cells entered microoxic conditions, the boundaries of the cyAbrB2 binding region and cyAbrB2-free region became obscure (Figure 5), “(l.319-320)

There is also no mention made anywhere about oligomerization potential for cyAbrB2,

We added the discussion about oligomerization “DNA-bound cyAbrB2 is expected to oligomerize, based on the long tract of cyAbrB2 binding region in our ChIP-seq data. However, no biochemical data mentioned the DNA deforming function or oligomerization of cyAbrB2 in the previous studies and preference for AT-rich DNA is not fully demonstrated in vitro (Dutheil et al. 2012; Ishii and Hihara 2008; Song et al. 2022)”(l. 277-280) and “Overall, the biochemistry integrating assay conditions (PTM, buffer condition, and cooperation with cyAbrB1) and output (DNA binding, oligomerization, and DNA structure) will deepen the understanding of cyAbrB2 as cyanobacterial NAPs.” (l.287-290)

The manuscript would benefit from some additional biochemical analyses of cyAbrB2 binding activity, with a particular focus on DNA binding and oligomerization/bridging

potential, and some additional discussion about these characteristics as well.

We added the discussion integrally considering known features of cyAbrB2, novel findings on cyAbrB2, and the comparison with known NAPs (ll.269-290).

(4) Given that the major take-away for the authors (based on the title) seems to be the nucleoid-associated protein potential for cyAbrB2, the Discussion would benefit from some additional focus in this area. How similar is cyAbrB2 to other nucleoid-associated proteins? (e.g. H-NS, Lsr2) How does counter-silencing work for other nucleoid-associated proteins? Can the authors definitively exclude the possibility of binding site competition/occlusion, given that cyAbrB2 covers the promoter region of hox? What is other nucleoid-associated proteins have been characterized in the cyanobacteria?

We agree with the point, so we additionally discussed cyAbrB2 comparing with H-NS and Lsr2, the canonical NAPs (ll. 269-290).

We did not deny the possibility of the exclusion of RNAP by cyAbrB2, but the previous manuscript insufficiently discussed that. To emphasize that cyAbrB2 excludes RNA polymerase, we simplified Figure 6 and employed mosaic plots showing anti-co-occurrence of cyAbrB2 binding regions and SigE peaks. Furthermore, we added discussion about SigE exclusion by cyAbrB2 (ll. 355-359)

We mention the possibility of other nucleoid-associated proteins in cyanobacteria in the discussion. “Furthermore, the conformational changes by deletion of cyAbrB2 were limited, suggesting there are potential NAPs in cyanobacteria yet to be characterized.” (ll.336-339)

(5) Previous work (Song et al., 2022) showed that changing the AT content of cyAbrB2 binding sites did not affect its ability to bind DNA. There are also previous papers suggesting that cyAbrB2 may be subject to diverse post-translational modifications (e.g. phosphorylation - Spat et al., 2023; glutathionylation - Sakr et al., 2013), as well as association with cyAbrB1. These collectively suggest there may be other factors that contribute to cyAbrB2 binding specificity/activity. These seem like relevant points to discuss, particularly given the transient nature of the cyAbrB2 effects on some genes.

We have included the discussion about AT content, post-translational modifications and transient regulations, and association with cyAbrB1 (ll. 284-295)

(6) Given the major binding site for SigA upstream of the hox operon, it seems that it likely also contributes to hox cluster expression, together with SigE. Is there a sense for the relative contribution of each sigma factor to hox cluster expression? And whether both are subject to the same inhibitory effect of cyAbrB2?

As described above response to the public review, the SigA binding site upstream of the hox operon should be assigned to the TSS of TU1715 (Figure 6C). Transcription of hox operon is highly dependent on SigE as shown in Figure S2, and residual transcription in sigEΔ strain is derived from other sigma factors (SigABCD). Estimating the relative contribution of sigma factors other than SigE is difficult at present because SigABCDE can partially compensate for each other.

As the different impact of NAPs on the primary and alternative sigma factor is observed in H-NS (Shin et al. 2005), whether both the primary sigma factor (SigA) and the alternative sigma factor (SigE) are inhibited by cyAbrB2 to the same extent is a very interesting question.

We calculated the odds ratio of SigE and SigA being in the cyAbrB2-free region and wrote in the result; “SigE preferred the cyAbrB2-free region in the aerobic condition more than SigA

did (Odds ratios of SigE and SigA being in the cyAbrB2-free region were 4.88 and 2.74, respectively).” (ll.193-195) and discussed “The higher exclusion pressure of cyAbrB2 on SigE may contribute to sharpening the transcriptional response of hox and nifJ on entry to microoxic conditions.” (ll.357-359)

(7) The 3C experiments suggest there are indeed changes in chromosome architecture in the hox region as growth conditions change and when different regulators are present. Across the chromosome, analogous changes are expected; however, it may be premature to draw this conclusion based on changes at one locus. Is there a reason that the authors did not take full advantage of their 3C samples and sequence them, to capture the full chromosome interactome at the two time-points? This would allow broader conclusions to be drawn regarding changes in chromosome structure and the impact of cyAbrB2.

In response to the suggestion, we performed an additional 3C assay on the nifJ region by utilizing residual 3C samples. Expanding to genome-wide sequence (Hi-C) needs concentration of ligated fragments by the biotinylation, which were omitted in our 3C sample.

We rewrote the result as obtained from the 3C data of hox and nifJ (ll.220-245) and omitted the schematic image of an entire chromosome of cyanobacteria (previous Figure 5E).

Editorial comments:

(1) The data presentation in Figure 1 is very effective.

(2) Line 87: please rephrase - you can have 'high similarity' or 'high levels of identity', but not high levels of homology - genes/proteins are either homologous or not.

(3) Line 118: classified into four 'groups'?

(4) Line 590: remove 'the'.

(5) Figure 2S, panel B: please define acronyms in the legend (GT, IP) and write out 'FLAG' in full for AbrB1.

(2) to (5) have been corrected.

(6) Please provide information on or a reference for the tagging of SigA for use in the ChIP-seq experiments within the Materials and Methods.

Added (l.365)

(7) Line 648: space between 'binding' and 'regions'.

corrected.

(8) Fig 4E: please make the solid lines thicker - they are currently difficult to see.

We have made Figure 6C (former 4E) larger and the line thicker.

(9) Line 666: location.

(10) Line 673: Individual.

(11) Figure S5, panel C graph title: should this be 'Relative'?

(12) Figure S7: What is 'GT'? Should this be 'WT'?

(9) to (12) have been corrected.

(13) In addition to the data presented in Figure 3G, it would be nice to have a small table or Venn diagram summarizing the number of cyAbrB2 binding sites that fall into the different categories (full gene/operon; downstream of a gene; within a gene; promoter region).

In response to the comment, we noticed the categories we had applied (full gene/operon; downstream of a gene; within a gene; promoter region) were arbitrary. Therefore, we categorized transcriptional units (TUs) according to the extent of occupancy by cyAbrB2. (Figures 4B and 4C)

(14) Line 280-281: suggest replacing 'mediates' with 'influences'. 'Mediates' sounds like a direct interaction (for which the evidence is not currently strong without some additional biochemical data), but 'influences' could better accommodate both direct and indirect possibilities.

(15) Line 410: it is not clear what this means.

We have omitted “As a result, DNA ~600-fold condensed DNA than 3C samples were ligated.”, as it does not give any information about the experimental procedure.

<https://doi.org/10.7554/eLife.94245.2.sa0>